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## DP

### D&C DP optimization

```
1 /**
2  Let opt(i, j) be the value of k that minimizes the above
3  expression. Assuming that the cost function satisfies
4  the quadrangle inequality, we can show that opt(i, j)
5  <= opt(i, j + 1) for all
6  i, j. This is known as the monotonicity condition. Then, we
7  can apply divide and conquer DP. The optimal "
8  splitting point" for a fixed i increases as j
9  increases.
10 */
11 int const N = 3e5 + 5, inf = 1e16;
12 int m, n, R = 0, L = 1, res = 0;
13 int ar[N], cnt[N];
14 vector<long long> dp_before, dp_cur;
15 void add(int id) {
16     res += cnt[ar[id]]; cnt[ar[id]]++;
17 }
18 void Remove(int id) {
```

```
13     cnt[ar[id]]--; res -= cnt[ar[id]];
14 }
15 long long C(int l, int r) {
16     while(R < r) add(++R);
17     while(L > l) add(--L);
18     while(L < l) Remove(L++);
19     while(R > r) Remove(R--);
20     return res;
21 }
22 void compute(int l, int r, int optl, int optpr) {
23     if (l > r) return;
24     int mid = (l + r) >> 1;
25     pair<long long, int> best = {LLONG_MAX, -1};
26     for (int k = optl; k <= min(mid, optpr); k++) best = min(
27         best, {(k ? dp_before[k - 1] : 0) + C(k, mid), k});
28     dp_cur[mid] = best.first;
29     int opt = best.second;
30     compute(l, mid - 1, optl, opt); compute(mid + 1, r, opt,
31         optpr);
32 }
33 void solve() {
34     cin >> n >> m;
35     for (int i = 1; i <= n; i++) cin >> ar[i];
36     dp_before.assign(n + 1, 0); dp_cur.assign(n + 1, 0);
37     for (int i = 1; i <= n; i++) dp_before[i] = C(1, i);
38     for (int i = 1; i < m; i++) {
39         compute(1, n, 1, n); dp_before = dp_cur;
40     }
41     cout << dp_before[n] << endl;
42 }
```

## SOS

```
1 const int B = 20;
2 int a[1 << B], f[1 << B], g[1 << B];
3 void solve() {
4     int n; cin >> n;
5     for (int i = 1; i <= n; i++) {
6         cin >> a[i];
7         f[a[i]]++;
8         g[a[i]]++;
9     }
10    // sum over subsets
11    for (int i = 0; i < B; i++) {
12        for (int mask = 0; mask < (1 << B); mask++) {
13            if ((mask & (1 << i)) != 0) {
14                f[mask] += f[mask ^ (1 << i)];
15            }
16        }
17    }
18    // sum over supersets
19    for (int i = 0; i < B; i++) {
20        for (int mask = (1 << B) - 1; mask >= 0; mask--) {
21            if ((mask & (1 << i)) == 0) g[mask] += g[mask ^ (1 <<
22                i)];
23        }
24    }
25    for (int i = 1; i <= n; i++) {
26        cout << f[a[i]] << ' ' << g[a[i]] << ' ' << n - f[(1 <<
27            B) - 1 - a[i]] << '\n';
28    }
```

## DS

## 2DSPARSE TABLE

```

1 const ll N=1000; const ll M=10; ll tab[M+1][M+1][N+1][N
+1]; ll L[N+1]; ll a[N+1][N+1]; ll s[N+1][N+1];
2 struct st2{ ll n,m; void init(ll _n,ll _m){ n=_n; m=_m; for
(ll i=2;i<=N;i++){ L[i]=L[i/2]+1; } } ll g(ll x,ll y,
ll z=0,ll w=0){ return max(x,y,z,w); } void build(){
for(ll i=0;i<n;i++){ for(ll j=0;j<m;j++){ tab[0][0][i
][j]=a[i][j]; } } for(ll y=1;y<=M;y++){ for(ll i=0;i<n
;i++){ for(ll j=0;j<m;j++){ if(j+(1<<y)-1<m) tab[0][y
][i][j]=g(tab[0][y-1][i][j],tab[0][y-1][i][j+(1<<(y-1
))]); } } } for(ll x=1;x<=M;x++){ for(ll i=0;i<n;i++){
for(ll j=0;j<m;j++){ if(i+(1<<x)-1<n) tab[x][0][i][j]=
g(tab[x-1][0][i][j],tab[x-1][0][i+(1<<(x-1))][j]); } }
for(ll x=1;x<=M;x++){ for(ll y=1;y<=M;y++){ for(ll i=0;i<n;i++){
for(ll j=0;j<m;j++){ if(i+(1<<x)-1<n &&
j+(1<<y)-1<m) { tab[x][y][i][j]=g( tab[x-1][y-1][i][j],
tab[x-1][y-1][i+(1<<(x-1))][j], tab[x-1][y-1][i][j
+(1<<(y-1))], tab[x-1][y-1][i+(1<<(x-1))][j+(1<<(y-1))
]); } } } } } ll qry_i(ll x,ll y,ll _x,ll _y){ ll
lx=L[_x-x+1]; ll ly=L[_y-y+1]; return g( tab[lx][ly][x
][y], tab[lx][ly][_x-(1<<lx)+1][y], tab[lx][ly][x][_y
-(1<<(ly))+1], tab[lx][ly][_x-(1<<lx)+1][_y-(1<<(ly))
+1]); } } };

```

## Alien Trick

```

1 // Aliens trick on Tree
2 // Maximizes total value using <= K selections
3 struct State {
4     long long val; // best value
5     int cnt; // number of selections
6 };
7 vector<pair<int,int>> g[N]; // {to, weight}
8 int K;
9 // merge two states
10 State merge(State a, State b) {
11     return {a.val + b.val, a.cnt + b.cnt};
12 }
13 State dfs(int u, int p, long long penalty) {
14     State res = {0, 0};
15     for (auto [v, w] : g[u]) {
16         if (v == p) continue;
17         State cur = dfs(v, u, penalty);
18         // choose edge
19         State take = {cur.val + w - penalty, cur.cnt + 1};
20         // choose better
21         if (take.val > cur.val) res = merge(res, take);
22         else res = merge(res, cur);
23     }
24     return res;
25 }
26 // check if we can do with <= K selections
27 bool check(long long penalty) {
28     State res = dfs(1, 0, penalty);
29     return res.cnt <= K;
30 }
31 // binary search on penalty
32 long long solve() {
33     long long lo = -1e12, hi = 1e12;
34     while (lo < hi) {
35         long long mid = (lo + hi + 1) / 2;

```

```

36         if (check(mid)) lo = mid;
37         else hi = mid - 1;
38     }
39     return dfs(1, 0, lo).val + lo * K;
40 }

```

## BIT

```

1 struct FenwickTree {
2     vector<int> bit; int n;
3     FenwickTree(int n) {
4         this->n = n; bit.assign(n, 0);
5     }
6     FenwickTree(vector<int> const& a) : FenwickTree(a.size())
7     {
8         for (size_t i = 0; i < a.size(); i++)
9             add(i, a[i]);
10    }
11    int sum(int r) {
12        int ret = 0;
13        for (; r >= 0; r = (r & (r + 1)) - 1)
14            ret += bit[r];
15        return ret;
16    }
17    int sum(int l, int r) {
18        return sum(r) - sum(l - 1);
19    }
20    void add(int idx, int delta) {
21        for (; idx < n; idx = idx | (idx + 1))
22            bit[idx] += delta;
23    }
24 }

```

## Centroid Decom

```

1 struct node {
2     vector<int> adj; int sub = 1; int p = -1; // centroid
3     int dep = 0; int vis = 0; // removed (centroid)
4 };
5 node tree[N]; int n; ll ans = 0; int root;
6 /* ----- SUBTREE SIZE ----- */
7 void dfs(int u, int p) {
8     tree[u].sub = 1;
9     for (int v : tree[u].adj) {
10        if (v != p && tree[v].vis == 0) {
11            dfs(v, u);
12            tree[u].sub += tree[v].sub;
13        }
14    }
15 }
16 /* ----- FIND CENTROID ----- */
17 int dfs2(int u, int p, int tot) {
18     for (int v : tree[u].adj) {
19         if (v != p && tree[v].vis == 0 && tree[v].sub > tot /
20             2) {
21             return dfs2(v, u, tot);
22         }
23     }
24     return u;
25 }
26 /* ----- COLLECT LEVEL INFO (OPTIONAL) ----- */
27 /* Keep this if you need depth / distance info */

```

```

27 void nextlev(int u, int p, int d) {
28     tree[u].dep = d;
29     for (int v : tree[u].adj) {
30         if (v != p && tree[v].vis == 0) {
31             nextlev(v, u, d + 1);
32         }
33     }
34 }
35 /* ----- SOLVE FOR A CENTROID ----- */
36 /*
37 // void solve(int c) {
38 // // example placeholder // do calculations using centroid
39 // }
40 /* ----- BUILD DECOMPOSITION ----- */
41 void build(int u, int p) {
42     dfs(u, -1);
43     int c = dfs2(u, -1, tree[u].sub);
44     tree[c].p = p;
45     if (p == -1) root = c;
46     // optional: gather depth info from centroid
47     nextlev(c, -1, 0);
48     solve(c);
49     tree[c].vis = 1;
50     for (int v : tree[c].adj) {
51         if (tree[v].vis == 0) {
52             build(v, c);
53         }
54     }
55 }
56 // Build the tree inside main and call build(1, -1);

```

## HASHING

```

1 int power(long long n, long long k, const int mod) { int
ans = 1 % mod; n %= mod; if (n < 0) n += mod; while (k
) { if (k & 1) ans = (long long) ans * n % mod; n = (
long long) n * n % mod; k >>= 1; } return ans; } const
int MOD1 = 127657753, MOD2 = 987654319; const int p1
= 137, p2 = 277; int ip1, ip2; pair<int, int> pw[N];
ipw[N]; void prec() { pw[0] = {1, 1}; for (int i = 1;
i < N; i++) { pw[i].first = 1LL * pw[i - 1].first * p1
% MOD1; pw[i].second = 1LL * pw[i - 1].second * p2 %
MOD2; } ip1 = power(p1, MOD1 - 2, MOD1); ip2 = power(
p2, MOD2 - 2, MOD2); ipw[0] = {1, 1}; for (int i = 1;
i < N; i++) { ipw[i].first = 1LL * ipw[i - 1].first *
ip1 % MOD1; ipw[i].second = 1LL * ipw[i - 1].second *
ip2 % MOD2; } } struct Hashing { int n; string s; // 0
- indexed vector<pair<int, int>> hs; // 1 - indexed
// Hashing() {} void init(string _s) { n = _s.size();
s = _s; hs.emplace_back(0, 0); for (int i = 0; i < n;
i++) { pair<int, int> p; p.first = (hs[i].first + 1LL
* pw[i].first * s[i] % MOD1 % MOD1; p.second = (hs[i
].second + 1LL * pw[i].second * s[i] % MOD2 % MOD2;
hs.push_back(p); } } pair<int, int> get_hash(int l,
int r) { // 1 - indexed assert(1 <= l && l <= r && r
<= n); pair<int, int> ans; ans.first = (hs[r].first -
hs[l - 1].first + MOD1) * 1LL * ipw[l - 1].first %
MOD1; ans.second = (hs[r].second - hs[l - 1].second +
MOD2) * 1LL * ipw[l - 1].second % MOD2; return ans; }
pair<int, int> get_hash() { return get_hash(1, n); }
};

```

## HLD and Merge sort tree

```

1 typedef struct node {
2     int p;
3     vector<int> adj;
4     int mxchild; int chain; int ind_in_chain; int subtree;
5     int koto; int price;
6 } node;
7 node tree[200010];
8 vector<int> chain(200010);
9 vector<int> start_c(200010);
10 vector<vector<int>> seg(800010);
11 vector<vector<ll>> sum(800010);
12 vector<int> dfs_array(200010);
13 // vector<vector<seg>> segtree(500010);
14 int c;
15 int ind=-1;
16 void dfs(int u, int p) {
17     tree[u].p = p;
18     int mx=0;
19     f(i, tree[u].adj.size()) {
20         int v = tree[u].adj[i];
21         if(p==v) continue;
22         dfs(v,u);
23         tree[u].subtree += tree[v].subtree;
24         if(mx < tree[v].subtree) {mx = tree[v].subtree; tree[u].mxchild = v;}
25     }
26 void hld(int u) {
27     dfs_array[++ind] = u;
28     chain[c]++;
29     // chain[c].push_back(u);
30     tree[u].chain = c;
31     tree[u].ind_in_chain = chain[c] - 1;
32     if(tree[u].mxchild) hld(tree[u].mxchild);
33     f(i, tree[u].adj.size()) {
34         int v = tree[u].adj[i];
35         if(v == tree[u].p || v == tree[u].mxchild) continue;
36         c++;
37         start_c[c] = ind+1;
38         hld(v);
39     }
40 }
41
42 void merge(int ind) {
43     int lsize = seg[2*ind+1].size();
44     int rsize = seg[2*ind+2].size();
45     seg[ind].clear();
46     sum[ind].clear();
47     seg[ind].resize(lsize+rsize);
48     sum[ind].resize(lsize+rsize);
49     int l = 0, r = 0, i=0;
50     while(l < lsize || r < rsize) {
51         if(l==lsize) {
52             seg[ind][i] = (seg[2*ind+2][r]);r++;
53         }
54         else if(r==rsize) {
55             seg[ind][i] = (seg[2*ind+1][l]);l++;
56         }
57         else if(seg[2*ind+1][l] < seg[2*ind+2][r]) {
58             seg[ind][i] = (seg[2*ind+1][l]);l++;
59         }
60         else {
61             seg[ind][i] = (seg[2*ind+2][r]);r++;
62         }

```

```

63         sum[ind][i] = ((l==0 ? 0 : sum[ind*2+1][l-1]) + (r==0
64             ? 0 : sum[ind*2+2][r-1]));
65         i++;
66         // return narr;
67     }
68 void build(vector<int>& arr, int l, int r, int ind) {
69     if(l==r) {
70         l = arr[l];
71         seg[ind].resize(1);
72         sum[ind].resize(1);
73         seg[ind][0] = (tree[l].price);
74         sum[ind][0] = ((ll)tree[l].koto);
75     } else {
76         int m = (l+r)/2;
77         build(arr,l,m,2*ind+1); build(arr,m+1,r,2*ind+2);
78         merge(ind);
79     }
80 }
81
82 ll query(int s, int e, int l, int r, int ind, int tar) {
83     if(s>e) return 0;
84     if(s==l && e==r) {
85         int ans = -1;
86         l = 0, r = seg[ind].size()-1;
87         while(l <= r) {
88             int m = (l+r)/2;
89             if(seg[ind][m] <= tar)
90                 ans = m, l = m+1;
91             else r = m-1;
92         }
93         return ans==-1?0:sum[ind][ans];
94     } else {
95         int m = (l+r)/2;
96         return query(s,min(e,m),l,m,2*ind+1,tar) + query(max(
97             s,m+1),e,m+1,r,2*ind+2,tar);
98     }

```

## HLD

```

1 vector<int> parent, depth, heavy, head, pos;
2 int cur_pos;
3 int dfs(int v, vector<vector<int>> const& adj) {
4     int size = 1; int max_c_size = 0;
5     for (int c : adj[v]) {
6         if (c != parent[v]) {
7             parent[c] = v, depth[c] = depth[v] + 1;
8             int c_size = dfs(c, adj); size += c_size;
9             if (c_size > max_c_size)
10                 max_c_size = c_size, heavy[v] = c;
11         }
12     }
13     return size;
14 }
15 void decompose(int v, int h, vector<vector<int>> const& adj) {
16     head[v] = h, pos[v] = cur_pos++;
17     if (heavy[v] != -1) decompose(heavy[v], h, adj);
18     for (int c : adj[v]) {
19         if (c != parent[v] && c != heavy[v]) decompose(c, c,
20             adj);
21     }
22 void init(vector<vector<int>> const& adj) {

```

```

23 int n = adj.size(); parent = vector<int>(n); depth =
24     vector<int>(n);
25 heavy = vector<int>(n, -1); head = vector<int>(n);
26 pos = vector<int>(n); cur_pos = 0;
27 dfs(0, adj);
28 decompose(0, 0, adj);
29 }

```

## LINE EQ

```

1 //all solution of ax+by=c in [minx,maxx], [miny,maxy]
2 int gcd(int a, int b, int &x, int &y) { if (b == 0) { x =
3     1; y = 0; return a; } int x1, y1; int d = gcd(b, a % b
4     , x1, y1); x = y1; y = x1 - y1 * (a / b); return d; }
5 bool solve_any(int a, int b, int c, int &x, int &y,
6     int &g) { if (a == 0 && b == 0) { g = x = y = 0;
7     return true; } g = gcd(abs(a), abs(b), x, y); if (c %
8     g != 0) { return false; } x *= (c / g); y *= (c / g);
9     if (a < 0) x = -x; if (b < 0) y = -y; return true; }
10 void shift(int &x, int &y, int a, int b, int cnt) { x
11     += cnt * b; y -= cnt * a; } int solve_all(int a, int b
12     , int c, int minx, int maxx, int miny, int maxy) { int
13     x, y, g; if (!solve_any(a, b, c, x, y, g)) return 0;
14     a /= g; b /= g; int sign_a = a > 0 ? +1 : -1; int
15     sign_b = b > 0 ? +1 : -1; shift(x, y, a, b, (minx - x)
16     / b); if (x < minx) { shift(x, y, a, b, sign_b); } if
17     (x > maxx) return 0; int lx1 = x; shift(x, y, a, b,
18     (maxx - x) / b); if (x > maxx) { shift(x, y, a, b, -
19     sign_b); } int rx1 = x; shift(x, y, a, b, (y - miny) /
20     a); if (y < miny) { shift(x, y, a, b, -sign_a); } if
21     (y > maxy) return 0; int lx2 = x; shift(x, y, a, b,
22     (y - maxy) / a); if (y > maxy) { shift(x, y, a, b,
23     sign_a); } int rx2 = x; if (lx2 > rx2) swap(lx2, rx2);
24     int lx = max(lx1, lx2); int rx = min(rx1, rx2); if (
25     lx > rx) return 0; return (rx - lx) / abs(b) + 1; }
26 void solve() { int a, b, c; cin >> a >> b >> c; int
27     minx, miny, maxx, maxy; cin >> minx >> maxx >> miny >>
28     maxy; if (maxx < minx || maxy < miny) { cout << 0 <<
29     endl; exit(0); } if (a == 0 && b == 0) { if (c == 0) {
30     cout << (maxy - miny + 1) * (maxx - minx + 1) << endl
31     ; } else { cout << 0 << endl; } } else if (a == 0 || b
32     == 0) { if (a == 0) { if (c % b == 0 && (c / b) >=
33     miny && (c / b) <= maxy) { cout << maxx - minx + 1 <<
34     endl; } else { cout << 0 << endl; } } else { if (c % a
35     == 0 && (c / a) >= minx && (c / a) <= maxx) { cout <<
36     maxy - miny + 1 << endl; } else { cout << 0 << endl;
37     } } } else cout << solve_all(a, b, c, minx, maxx, miny
38     , maxy) << endl; }

```

## MOs

```

1 const int BLOCK_SIZE = 350;
2 int n, q, ar[N], freq[N], res = 0, ans[10*N];
3 struct Query {
4     int l, r, id;
5     bool operator<(const Query &y) const {
6         int x_block = l / BLOCK_SIZE;
7         int y_block = y.l / BLOCK_SIZE;
8         if(x_block == y_block)
9             return r < y.r;
10        return x_block < y_block;
11    }
12 };

```

```

13 void add(int i) {
14     int x = ar[i]; res -= freq[x] / 2; freq[x]++; res +=
        freq[x] / 2;
15 }
16 void Remove(int i) {
17     int x = ar[i]; res -= freq[x] / 2; freq[x]--; res +=
        freq[x] / 2;
18 }
19 void solve() { // need to reset freq if needed
20     cin >> n;
21     for (int i = 1; i <= n; i++) cin >> ar[i];
22     cin >> q;
23     vector<Query> qr;
24     res = 0; int ii = 0;
25     while(q--) {
26         int l, r; cin >> l >> r; qr.push_back(Query({l, r, ii
            ++}));
27     }
28     sort(qr.begin(), qr.end());
29     int left = 1, right = 0;
30     for (Query q : qr) {
31         int l = q.l, r = q.r;
32         while(right < r) add(++right);
33         while(left > l) add(--left);
34         while(left < l) Remove(left++);
35
36         while(right > r) Remove(right--);
37         ans[q.id] = res;
38     }
39     for (int i = 0; i < ii; i++) cout << ans[i] << "\n";
40 }

```

## Persistence trie

```

1 map<ll,ll>root; struct node { ll left = 0, right = 0; ll
    cnt = 0; }; node a[9000001]; ll ptr=2; struct p_trie{
    p_trie() { root[0] = 1; } void insert(ll par_id,ll
    curr_id,ll k){ root[curr_id]=ptr++; ll par_node=root[
    par_id]; ll curr_node=root[curr_id]; for(ll i=30;i>=0;
    i--){ if((k&(1ll<<i))) { a[curr_node].left = a[
    par_node].left; a[curr_node].right = ptr++; curr_node
    = a[curr_node].right; par_node = a[par_node].right; }
    else{ a[curr_node].right=a[par_node].right; a[
    curr_node].left=ptr++; curr_node=a[curr_node].left;
    par_node=a[par_node].left; } a[curr_node].cnt=a[
    par_node].cnt+1; } } ll max_xor(ll id,ll k){ ll curr=
    root[id]; ll ans=0; for(ll i=30;i>=0;i--){ if((k&(1ll
    <<i))){ if(a[curr].left){ ans|=(1ll<<i); curr=a[curr].
    left; }else{ curr=a[curr].right; } }else{ if(a[curr].
    right){ ans|=(1ll<<i); curr=a[curr].right; }else{ curr
    =a[curr].left; } } } return ans; } ll min_xor(ll id,ll
    k){ ll curr=root[id]; ll ans=0; for(ll i=30;i>=0;i--
    ){ if((k&(1ll<<i))){ if(a[curr].right){ curr=a[curr].
    right; }else{ ans|=(1ll<<i); curr=a[curr].left; } }
    else{ if(a[curr].left){ curr=a[curr].left; }else{ ans
    |= (1ll<<i); curr=a[curr].right; } } } return ans; } }

```

## Trie

```

1 const int N = 3e5 + 9;
2
3 struct Trie {
4     static const int B = 31;

```

```

5
6 struct node {
7     node* nxt[2];
8     int sz;
9
10    node() {
11        nxt[0] = nxt[1] = NULL;
12        sz = 0;
13    }
14 } *root;
15 Trie() {
16     root = new node();
17 }
18 void insert(int val) {
19     node* cur = root;
20     cur->sz++;
21     for (int i = B - 1; i >= 0; i--) {
22         int b = (val >> i) & 1;
23         if (cur->nxt[b] == NULL)
24             cur->nxt[b] = new node();
25         cur = cur->nxt[b];
26         cur->sz++;
27     }
28 }
29 void remove(int val) {
30     node* cur = root;
31     cur->sz--;
32
33     for (int i = B - 1; i >= 0; i--) {
34         int b = (val >> i) & 1;
35
36         cur = cur->nxt[b];
37         cur->sz--;
38     }
39 }
40 int get_min(int x) {
41     node* cur = root;
42     int ans = 0;
43
44     for (int i = B - 1; i >= 0; i--) {
45         int b = (x >> i) & 1;
46         if (cur->nxt[b] && cur->nxt[b]->sz > 0) {
47             cur = cur->nxt[b];
48         }
49         else {
50             cur = cur->nxt[!b];
51             ans |= (1LL << i);
52         }
53     }
54
55     return ans;
56 }
57
58 int get_max(int x) {
59     node* cur = root;
60     int ans = 0;
61
62     for (int i = B - 1; i >= 0; i--) {
63         int b = (x >> i) & 1;
64         if (cur->nxt[!b] && cur->nxt[!b]->sz > 0) {
65             cur = cur->nxt[!b];
66             ans |= (1LL << i);
67         }
68         else {
69             cur = cur->nxt[b];

```

```

70     }
71 }
72 return ans;
73 }
74 int query(int x, int k) {
75     node* cur = root;
76     int ans = 0;
77
78     for (int i = B - 1; i >= 0; i--) {
79         if (cur == NULL) break;
80
81         int b1 = (x >> i) & 1;
82         int b2 = (k >> i) & 1;
83
84         if (b2 == 1) {
85             if (cur->nxt[b1])
86                 ans += cur->nxt[b1]->sz;
87
88             cur = cur->nxt[!b1];
89         }
90         else {
91             cur = cur->nxt[b1];
92         }
93     }
94     return ans;
95 }
96 void del(node* cur) {
97     for (int i = 0; i < 2; i++) {
98         if (cur->nxt[i])
99             del(cur->nxt[i]);
100     }
101     delete cur;
102 }
103 ~Trie() {
104     del(root);
105 }
106 } t;

```

## persistentSegment

```

1 struct node {
2     ll sum; node *left, *right;
3     node(ll val = 0) {
4         sum = val; left = right = NULL;
5     }
6     node(node *l, node *r) {
7         left = l; right = r; sum = (l ? l->sum : 0) + (r ? r
            ->sum : 0);
8     }
9 };
10 int n; vector<int> v;
11 node* build(int start, int end) {
12     if (start == end)
13         return new node(v[start]);
14     int mid = (start + end) >> 1;
15     return new node(build(start, mid), build(mid + 1, end));
16 }
17 node* update(node* prev, int pos, ll val, int start, int
    end) {
18     if (start == end) return new node(val);
19     int mid = (start + end) >> 1;
20     if (pos <= mid) return new node(update(prev->left, pos,
        val, start, mid), prev->right);
21     else return new node(prev->left, update(prev->right, pos
        , val, mid + 1, end));

```

```

22 }
23 ll query(node* cur, int l, int r, int start, int end) {
24     if (cur == NULL || start > r || end < l) return 0;
25     if (l <= start && end <= r) return cur->sum;
26     int mid = (start + end) >> 1;
27     return query(cur->left, l, r, start, mid) + query(cur->
        right, l, r, mid + 1, end);
28 }
29 void solve() {
30     cin >> n; v.resize(n);
31     for (int &x : v) cin >> x;
32     vector<node*> roots; roots.push_back(build(0, n - 1));
33     int q; cin >> q;
34     while (q--) {
35         int id; cin >> id;
36         if (id == 1) {
37             int ver, pos; ll val; cin >> ver >> pos >> val;
38             roots.push_back(update(roots[ver], pos - 1, val,
                0, n - 1));
39         } else {
40             int ver, l, r; cin >> ver >> l >> r;
41             cout << query(roots[ver], l - 1, r - 1, 0, n - 1)
                << "\n";
42         }
43     }
44 }

```

## randomnumber

```

1 struct custom_hash {
2     static uint64_t splitmix64(uint64_t x) {
3         x += 0x9e3779b97f4a7c15;
4         x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
5         x = (x ^ (x >> 27)) * 0x94d049b133111eb;
6         return x ^ (x >> 31);
7     }
8     size_t operator()(uint64_t x) const {
9         static const uint64_t FIXED_RANDOM = chrono::
10             steady_clock::now().time_since_epoch().count();
11         return splitmix64(x + FIXED_RANDOM);}; //For long long int
            to long long int

```

## xor\_basis

```

1 struct XorBasis {
2     vector<ll> basis;
3     ll N = 0, tmp = 0;
4     void add(ll x) {
5         N++;
6         tmp ^= x;
7         for (ll b : basis) x = min(x, x ^ b);
8         if (!x) return;
9         for (ll &b : basis) if ((b ^ x) < b) b ^= x;
10        basis.push_back(x);
11        sort(basis.rbegin(), basis.rend());
12    }
13    // number of basis vectors (rank)
14    ll size() { return (ll) basis.size(); }
15    // clear everything
16    void clear() { N = 0; tmp = 0; basis.clear(); }
17    // returns true if x can be formed
18    bool possible(ll x) {
19        for (ll b : basis) x = min(x, x ^ b);

```

```

20     return x == 0;
21 }
22 // returns maximum xor value
23 ll maxxor(ll x = 0) {
24     for (ll b : basis) x = max(x, x ^ b);
25     return x;
26 }
27 // returns minimum xor value
28 ll minxor(ll x = 0) {
29     for (ll b : basis) x = min(x, x ^ b);
30     return x;
31 }
32 // returns count of subsets which xor to x (modded)
33 ll cntxor(ll x) {
34     if (!possible(x)) return 0LL;
35     ll freebits = N - size();
36     ll res = 1;
37     while (freebits--) res = (res * 2) % MOD;
38     return res;
39 }
40 // returns sum of xor of all subsets
41 ll sumOfAll() {
42     return tmp * (1LL << (N - 1));
43 }
44 // returns k-th smallest xor value (1-indexed)
45 ll kth(ll k) {
46     ll sz = size();
47     if (k > (1LL << sz)) return -1;
48     k--;
49     ll ans = 0;
50     for (ll i = 0; i < sz; i++)
51         if (k >> (sz - 1 - i) & 1) ans ^= basis[i];
52     return ans;
53 }
54 } xb;

```

## Flow

### Dinic

```

1 #include<bits/stdc++.h>
2 using namespace std;
3
4 const int N = 5010;
5
6 const long long inf = 1LL << 61;
7 struct Dinic {
8     struct edge {
9         int to, rev;
10        long long flow, w;
11        int id;
12    };
13    int n, s, t, mxid;
14    vector<int> d, flow_through;
15    vector<int> done;
16    vector<vector<edge>> g;
17    Dinic() {}
18    Dinic(int _n) {
19        n = _n + 10;
20        mxid = 0;
21        g.resize(n);
22    }
23    void add_edge(int u, int v, long long w, int id = -1) {
24        edge a = {v, (int)g[v].size(), 0, w, id};

```

```

25        edge b = {u, (int)g[u].size(), 0, 0, -2}; //for
            bidirectional edges cap(b) = w
26        g[u].emplace_back(a);
27        g[v].emplace_back(b);
28        mxid = max(mxid, id);
29    }
30    bool bfs() {
31        d.assign(n, -1);
32        d[s] = 0;
33        queue<int> q;
34        q.push(s);
35        while (!q.empty()) {
36            int u = q.front();
37            q.pop();
38            for (auto &e : g[u]) {
39                int v = e.to;
40                if (d[v] == -1 && e.flow < e.w) d[v] = d[u] + 1, q.
                    push(v);
41            }
42        }
43        return d[t] != -1;
44    }
45    long long dfs(int u, long long flow) {
46        if (u == t) return flow;
47        for (int &i = done[u]; i < (int)g[u].size(); i++) {
48            edge &e = g[u][i];
49            if (e.w <= e.flow) continue;
50            int v = e.to;
51            if (d[v] == d[u] + 1) {
52                long long nw = dfs(v, min(flow, e.w - e.flow));
53                if (nw > 0) {
54                    e.flow += nw;
55                    g[v][e.rev].flow -= nw;
56                    return nw;
57                }
58            }
59        }
60        return 0;
61    }
62    long long max_flow(int _s, int _t) {
63        s = _s;
64        t = _t;
65        long long flow = 0;
66        while (bfs()) {
67            done.assign(n, 0);
68            while (long long nw = dfs(s, inf)) flow += nw;
69        }
70        flow_through.assign(mxid + 10, 0);
71        for (int i = 0; i < n; i++) for (auto e : g[i]) if (e.id >=
            0) flow_through[e.id] = e.flow;
72        return flow;
73    }
74 };
75 int main() {
76     int n, m;
77     cin >> n >> m;
78     Dinic F(n + 1);
79     for (int i = 1; i <= m; i++) {
80         int u, v, w;
81         cin >> u >> v >> w;
82         F.add_edge(u, v, w);
83     }
84     cout << F.max_flow(1, n) << "\n";
85     return 0;
86 }

```

## edmondskarp

```

1 // edmonds karp ve^2 call while(bfs(n) != 0)
2 typedef struct node {
3     int vis; int p; vector<int> adj; vector<ll> cap;
4 } node;
5 node graph[519];
6 ll bfs(int n) {
7     queue<int> qu;
8     vector<bool> vis(510,0);
9     vis[1] = 1;
10    qu.push(1);
11    while(!qu.empty()) {
12        int cur = qu.front();
13        // printf("cur-- %d\n", cur);
14        qu.pop();
15        f(i, graph[cur].adj.size()) {
16            int nex=graph[cur].adj[i];
17            if(vis[nex] == 0 && graph[cur].cap[nex] > 0) {
18                graph[nex].p = cur; vis[nex] = 1;
19                if(nex==n) break;
20                qu.push(nex);
21            }
22        }
23    }
24    if(vis[n] == 0) return 0;
25    ll maxflow = LLONG_MAX;
26    int cur = n;
27    while(cur != 1) {
28        maxflow = min(maxflow, graph[graph[cur].p].cap[cur]);
29        cur = graph[cur].p;
30        // printf("%d ", cur);
31    }
32    // printf("max %d\n", maxflow);
33    cur = n;
34    while(cur != 1) {
35        graph[graph[cur].p].cap[cur] -= maxflow;
36        graph[cur].cap[graph[cur].p] += maxflow;
37        cur = graph[cur].p;
38    }
39    return maxflow;
40 }
41
42 // call after getting max flow
43 vector<int> getCut(int n) {
44     vector<int> vis(510,0);
45     queue<int> q;
46     q.push(1);
47     vis[1] = 1;
48     while(!q.empty()) {
49         int u = q.front(); q.pop();
50         for(int v : graph[u].adj) {
51             if(!vis[v] && graph[u].cap[v] > 0) { // residual
52                 capacity
53                 vis[v] = 1;
54                 q.push(v);
55             }
56         }
57     }
58     return vis; // vis[v] = 1 -> reachable -> S-side

```

## mcmf

```

1 struct Edge
2 {
3     int from, to, capacity, cost;
4 };
5 vector<vector<int>> adj, cost, capacity;
6 const int INF = 1e9;
7 void shortest_paths(int n, int v0, vector<int>& d, vector<
8     int>& p) {
9     d.assign(n, INF);
10    d[v0] = 0;
11    vector<bool> inq(n, false);
12    queue<int> q;
13    q.push(v0);
14    p.assign(n, -1);
15    while (!q.empty()) {
16        int u = q.front();
17        q.pop();
18        inq[u] = false;
19        for (int v : adj[u]) {
20            if (capacity[u][v] > 0 && d[v] > d[u] + cost[u][v]) {
21                d[v] = d[u] + cost[u][v];
22                p[v] = u;
23                if (!inq[v]) {
24                    inq[v] = true;
25                    q.push(v);
26                }
27            }
28        }
29    }
30 }
31 int min_cost_flow(int N, vector<Edge> edges, int K, int s,
32     int t) {
33     adj.assign(N, vector<int>());
34     cost.assign(N, vector<int>(N, 0));
35     capacity.assign(N, vector<int>(N, 0));
36     for (Edge e : edges) {
37         adj[e.from].push_back(e.to);
38         adj[e.to].push_back(e.from);
39         cost[e.from][e.to] = e.cost;
40         cost[e.to][e.from] = -e.cost;
41         capacity[e.from][e.to] = e.capacity;
42     }
43     int flow = 0;
44     int cost = 0;
45     vector<int> d, p;
46     while (flow < K) {
47         shortest_paths(N, s, d, p);
48         if (d[t] == INF) break;
49         // find max flow on that path
50         int f = K - flow;
51         int cur = t;
52         while (cur != s) {
53             f = min(f, capacity[p[cur]][cur]);
54             cur = p[cur];
55         }
56         // apply flow
57         flow += f;
58         cost += f * d[t];
59         cur = t;
60         while (cur != s) {
61             capacity[p[cur]][cur] -= f;
62             capacity[cur][p[cur]] += f;
63             cur = p[cur];

```

```

63     }
64 }
65 if (flow < K)
66     return -1;
67 else
68     return cost;
69 }

```

## General formulas

```

1 /* ----- POLYGON -----
2 Area of polygon (shoelace):
3 Area = 0.5 * |sum(x[i]*y[i+1] - x[i+1]*y[i])|
4 Centroid of polygon:
5 Cx = sum((xi + xi+1) * cross(i,i+1)) / (6*Area)
6 Cy = sum((yi + yi+1) * cross(i,i+1)) / (6*Area)
7 Point inside polygon (ray casting):
8 Count intersections of ray to +x
9 Odd = inside, Even = outside
10 Convex polygon point inside:
11 For all edges (Pi, Pi+1):
12 cross(Pi+1-Pi, X-Pi) >= 0 (or <=0 consistently)
13 */
14 /* ----- TRIANGLE -----
15 Area using cross:
16 Area = |cross(B-A, C-A)| / 2
17 Circumcenter:
18 Intersection of perpendicular bisectors
19 (Omitted formula compute via lines)
20 Incenter:
21 I = (a*A + b*B + c*C) / (a+b+c)
22 where:
23 Centroid:
24 G = (A+B+C)/3
25 Orthocenter:
26 H = A + B + C - 2*O (O = circumcenter)
27 Radius:
28 Inradius r = Area / s
29 Circumradius R = (a*b*c) / (4*Area)
30 Angle between vectors:
31 cos(theta) = dot(u,v)/(|u||v|)
32 */
33 /* ----- COMBINATORICS & MATH -----
34 Stirling Numbers of the Second Kind S(n,k) Parity:
35 Odd = ((n - k) & ((k - 1) / 2)) == 0)
36 Even = otherwise
37
38 Sum of Powers (Faulhaber's Formula):
39 // Based on image_362f5e.png
40 sum(k^p) for k=1..n = (1 / (p + 1)) * sum_{r=0 to p} [ C(p
41     +1, r) * B_r * n^(p+1-r) ]
42 * Note: B_r represents the r-th Bernoulli number (using B_1
43     ^+ = +1/2).
44
45 Bracket Arrangements:
46 // DP[i][j]: ways to arrange i pairs of brackets into j
47     valid non-empty blocks
48 DP[i][j] = (j * C(2i - j, i - j) * inv(2i - j)) % M
49 */

```

## generate\_codebook

```

1 import os
2 import subprocess
3 import sys
4
5 ALLOWED_EXTENSIONS = {
6     ".cpp": "C++",
7     ".cc": "C++",
8     ".h": "C++",
9     ".hpp": "C++",
10    ".py": "Python",
11    ".java": "Java",
12 }
13
14 IGNORE_DIRS = {".git", ".github", "build", "__pycache__"}
15
16 def escape_latex(text: str) -> str:
17     replacements = {
18         "\\": r"\textbackslash{}",
19         "_": r"\_",
20         "%": r"\%",
21         "$": r"\$",
22         "#": r"\#",
23         "&": r"\&",
24         "{": r"\{",
25         "}": r"\}",
26         "~": r"\textasciitilde{}",
27         "^": r"\textasciicircum{}",
28     }
29     for char, rep in replacements.items():
30         text = text.replace(char, rep)
31     return text
32
33 def collect_files():
34     tree = {}
35     for root, dirs, files in os.walk("."):
36         dirs[:] = [d for d in dirs if d not in IGNORE_DIRS
37                    and not d.startswith(".")]
38         rel_dir = os.path.relpath(root, ".")
39         valid_files = [
40             f for f in sorted(files) if os.path.splitext(f)[1]
41             in ALLOWED_EXTENSIONS
42         ]
43         if valid_files:
44             section_name = "General" if rel_dir == "." else
45             rel_dir
46             tree[section_name] = [
47                 os.path.join(root, f).replace("\\", "/") for f
48                 in valid_files
49             ]
50     return tree
51
52 def build_latex_document(tree, is_color: bool) -> str:
53     color_definitions = (
54         r"""
55 \definecolor{kwcolor}{rgb}{0.0, 0.2, 0.8}
56 \definecolor{cmtcolor}{rgb}{0.0, 0.5, 0.0}
57 \definecolor{strcolor}{rgb}{0.7, 0.1, 0.1}
58 \definecolor{numcolor}{rgb}{0.4, 0.4, 0.4}
59 \definecolor{framecolor}{rgb}{0.7, 0.7, 0.7}
60 """
61         if is_color
62     else r"""
63 \definecolor{kwcolor}{rgb}{0.0, 0.0, 0.0}
64 \definecolor{cmtcolor}{rgb}{0.2, 0.2, 0.2}
65 \definecolor{strcolor}{rgb}{0.0, 0.0, 0.0}
66 \definecolor{numcolor}{rgb}{0.3, 0.3, 0.3}
67 \definecolor{framecolor}{rgb}{0.0, 0.0, 0.0}
68 """
69     )
70     doc = [
71         r"\documentclass[9pt,landscape,a4paper]{article}",
72         r"\usepackage[left=0.8cm,right=0.8cm,top=1.2cm,bottom=1.2cm]{geometry}",
73         r"\usepackage{multicol}",
74         r"\usepackage{listings}",
75         r"\usepackage{xcolor}",
76         r"\usepackage{fancyhdr}",
77         r"\usepackage{titlesec}",
78         r"\usepackage{tocloft}",
79         r"\usepackage{courier}",
80         color_definitions,
81         r"\lstset{",
82         r"    basicstyle=\ttfamily\scriptsize,",
83         r"    keywordstyle=\bfseries\color{kwcolor},",
84         r"    commentstyle=\itshape\color{cmtcolor},",
85         r"    stringstyle=\color{strcolor},",
86         r"    numberstyle=\tiny\color{numcolor},",
87         r"    numbers=left,",
88         r"    stepnumber=1,",
89         r"    numbersep=4pt,",
90         r"    frame=single,",
91         r"    rulecolor=\color{framecolor},",
92         r"    tabsize=2,",
93         r"    breaklines=true,",
94         r"    breakatwhitespace=false,",
95         r"    showstringspaces=false,",
96         r"    columns=fullflexible,",
97         r"    xleftmargin=12pt,",
98         r"}",
99         r"\pagestyle{fancy}",
100        r"\fancyhf{}",
101        r"\fancyhead[L]{\textbf{Algorithmic Code Library}}",
102        (
103            r"\fancyhead[C]{\textit{Color Edition}}"
104            if is_color
105            else r"\fancyhead[C]{\textit{B\&W Edition}}"
106        ),
107        r"\fancyhead[R]{\thepage}",
108        r"\renewcommand{\headrulewidth}{0.4pt}",
109        r"\titlespacing*{\section}{0pt}{4pt}{2pt}",
110        r"\titlespacing*{\subsection}{0pt}{3pt}{1pt}",
111        r"\begin{document}",
112        r"\begin{multicols*}{3}",
113        r"\tableofcontents",
114        r"\vspace{10pt}",
115        r"\hrule",
116        r"\vspace{10pt}",
117    ]
118
119    for section, files in sorted(tree.items()):
120        doc.append(f"\section*{{{escape_latex(section)}}}")
121        doc.append(f"\addcontentsline{{toc}}{{section}}{{{escape_latex(section)}}}")
122
123    for file_path in files:
124        file_name = os.path.basename(file_path)
125        title = os.path.splitext(file_name)[0]
126        ext = os.path.splitext(file_name)[1]
127        lang = ALLOWED_EXTENSIONS.get(ext, "C++")
128
129        doc.append(f"\subsection*{{{escape_latex(title)}}}")
130        doc.append(
131            f"\addcontentsline{{toc}}{{subsection}}{{{escape_latex(title)}}}"
132        )
133        doc.append(f"\lstinputlisting[language={lang}]{{{file_path}}}")
134
135    doc.append(r"\end{multicols*}")
136    doc.append(r"\end{document}")
137    return "\n".join(doc)
138
139 def compile_latex(tex_name: str):
140     # Run twice so table of contents page numbers compute correctly
141     for _ in range(2):
142         cmd = ["pdflatex", "-interaction=nonstopmode", tex_name]
143         # We capture the output but don't immediately crash on a non-zero return code
144         subprocess.run(cmd, stdout=subprocess.PIPE, stderr=subprocess.PIPE)
145
146     # Instead of checking the strict return code, check if the PDF actually generated
147     pdf_name = tex_name.replace(".tex", ".pdf")
148     if not os.path.exists(pdf_name):
149         print(f"Error: {pdf_name} was not generated!")
150         sys.exit(1)
151
152 def main():
153     tree = collect_files()
154     if not tree:
155         print("No source code files found to compile.")
156         sys.exit(1)
157
158     # 1. Color Edition
159     with open("codebook_color.tex", "w", encoding="utf-8") as f:
160         f.write(build_latex_document(tree, is_color=True))
161         compile_latex("codebook_color.tex")
162
163     # 2. Black & White Edition
164     with open("codebook_bw.tex", "w", encoding="utf-8") as f:
165         f.write(build_latex_document(tree, is_color=False))
166         compile_latex("codebook_bw.tex")
167
168     print("Successfully built codebook_color.pdf and codebook_bw.pdf")
169
170 if __name__ == "__main__":
171     main()

```

sublime build

```

1 {
2   "cmd" : ["g++ -std=c++20 $file_name -o $file_base_name &&
           timeout 5s ./ $file_base_name<input.txt>output.txt"
           ],
3   "selector" : "source.cpp",
4   "shell": true
5   //"working_dir" : "$file_path"
6 }

```

## template

```

1 #include<bits/stdc++.h>
2 using namespace std;
3 #define endl '\n'
4 #include <ext/pb_ds/assoc_container.hpp>
5 #include <ext/pb_ds/tree_policy.hpp>
6 using namespace __gnu_pbds;
7 #define int long long
8 typedef tree<int ,null_type,less<int >,rb_tree_tag,
9 tree_order_statistics_node_update> o_set;
10 typedef tree<int ,null_type,less_equal<int>,rb_tree_tag,
11 tree_order_statistics_node_update> o_multiset;
12 mt19937 rng(chrono::steady_clock::now().time_since_epoch().
13 count());
14 void solve(){ //order_of_key, find_by_order }
15 int32_t main() {
16   ios::sync_with_stdio(false);cin.tie(nullptr);
17   int tc = 1; cin >> tc;
18   for(int i = 1; i <= tc; i++) { solve(); } return 0;
19 }

```

## Geometry

### Convex hull

```

1 vector<PT> convex_hull(vector<PT> &p) {
2   if (p.size() <= 1) return p;
3   vector<PT> v = p;
4   sort(v.begin(), v.end());
5   vector<PT> up, dn;
6   for (auto& p : v) {
7     while (up.size() > 1 && orientation(up[up.size() -
8     2], up.back(), p) >= 0) {
9       up.pop_back();
10    }
11    while (dn.size() > 1 && orientation(dn[dn.size() -
12    2], dn.back(), p) <= 0) {
13      dn.pop_back();
14    }
15    up.push_back(p);
16    dn.push_back(p);
17  }
18  v = dn;
19  if (v.size() > 1) v.pop_back();
20  reverse(up.begin(), up.end());
21  up.pop_back();
22  for (auto& p : up) {
23    v.push_back(p);
24  }
25  if (v.size() == 2 && v[0] == v[1]) v.pop_back();
26  return v;

```

```

26 //checks if convex or not
27 bool is_convex(vector<PT> &p) {
28   bool s[3]; s[0] = s[1] = s[2] = 0;
29   int n = p.size();
30   for (int i = 0; i < n; i++) {
31     int j = (i + 1) % n;
32     int k = (j + 1) % n;
33     s[sign(cross(p[j] - p[i], p[k] - p[i])) + 1] = 1;
34     if (s[0] && s[2]) return 0;
35   }
36   return 1;
37 }

```

### Geo 2

```

1 struct point { int x, y, idx; }; //Finds squared euclidean
2 distance between two points int dist(point &a, point &
3 b) { return (a.x-b.x)*(a.x-b.x) + (a.y-b.y)*(a.y-b.y); }
4 //Checks if angle ABC is a right angle int
5 isOrthogonal(point &a, point &b, point &c) { return (b
6 .x-a.x) * (b.x-c.x) + (b.y-a.y) * (b.y-c.y) == 0; } //
7 Checks if ABCD form a rectangle (in that order) int
8 isRectangle(point &a, point &b, point &c, point &d) {
9 return isOrthogonal(a, b, c) && isOrthogonal(b, c, d)
10 && isOrthogonal(c, d, a); } //Checks if ABCD form a
11 rectangle, in any orientation int isRectangleAnyOrder(
12 point &a, point &b, point &c, point &d) { return
13 isRectangle(a, b, c, d) || isRectangle(b, c, a, d) |
14 isRectangle(c, a, b, d); } //Checks if ABCD form a
15 square (in that order) int isSquare(point &a, point &b
16 , point &c, point &d) { return isRectangle(a, b, c, d)
17 && dist(a, b) == dist(b, c); } //Checks if ABCD form
18 a square, in any orientation int isSquareAnyOrder(
19 point &a, point &b, point &c, point &d) { return
20 isSquare(a, b, c, d) || isSquare(b, c, a, d) |
21 isSquare(c, a, b, d); }

```

### Nearest point

```

1 const int N = 1<<17; const long long INF =
2 2000000000000000000; struct vp_node { long long
3 threshold; pair< int, int > center; vp_node *left, *
4 right; }; typedef vp_node *node_ptr; int tests,n; pair
5 < int, int > pts[N],arr[N]; long long ans; node_ptr
6 root; long long squared_distance(pair< int, int > a,
7 pair< int, int > b) { return (a.first-b.first)*1ll*(a
8 .first-b.first) + (a.second-b.second)*1ll*(a.second-b
9 .second); } struct cmp_points { pair< int, int > a;
10 cmp_points(){} cmp_points(pair< int, int > p): a(p)
11 {} bool operator () (const pair< int, int > &x, const
12 pair< int, int > &y) const { return squared_distance
13 (a,x)<squared_distance(a,y); }; void build(node_ptr
14 &node, int a, int b) { if(a>b) { node=NULL; return; }
15 node=new vp_node(); if(a==b) { node->threshold=0; node
16 ->center=arr[a]; node->left=NULL; node->right=NULL;
17 return; } int pos=a+rand()% (b-a+1); swap(arr[a],arr[
18 pos]); node->center=arr[a]; sort(arr+a+1,arr+b+1,
19 cmp_points(arr[a])); node->threshold=squared_distance(
20 arr[a],arr[(a+b)>>1]); build(node->left,a,(a+b)>>1);
21 build(node->right,((a+b)>>1)+1,b); } void query(
22 node_ptr &node, pair< int, int > q, long long &ans) {
23 if(node==NULL) return; if(node->center!=q) ans=min(
24 ans,squared_distance(node->center,q)); if(node->left==

```

```

1 NULL && node->right==NULL) { return; } if(
2 squared_distance(q,node->center)<=node->threshold) {
3 query(node->left,q,ans); if(sqrt(squared_distance(node
4 ->center,q))+sqrt(ans)>sqrt(node->threshold)) query(
5 node->right,q,ans); } else { query(node->right,q,ans);
6 if(sqrt(squared_distance(node->center,q))-sqrt(ans)<
7 sqrt(node->threshold)) query(node->left,q,ans); } }
8 int main() { int i; scanf("%d", &tests); while(tests
9 --) { scanf("%d", &n); for(i=1;i<=n;i++) { scanf("%d %
10 d", &arr[i].first, &arr[i].second); pts[i]=arr[i]; }
11 build(root,1,n); for(i=1;i<=n;i++) { ans=INF; query(
12 root,pts[i],ans); printf("%lld\n", ans); } } return 0;
13 }

```

### geo\_2d

```

1 const double PI = acos((double)-1.0); int sign(double x) {
2 return (x > eps) - (x < -eps); } struct PT { double x,
3 y; PT() { x = 0, y = 0; } PT(double x, double y) : x(
4 x), y(y) {} PT(const PT &p) : x(p.x), y(p.y) {} PT
5 operator + (const PT &a) const { return PT(x + a.x, y
6 + a.y); } PT operator - (const PT &a) const { return
7 PT(x - a.x, y - a.y); } PT operator * (const double a)
8 const { return PT(x * a, y * a); } friend PT operator
9 * (const double &a, const PT &b) { return PT(a * b.x,
10 a * b.y); } PT operator / (const double a) const {
11 return PT(x / a, y / a); } bool operator == (PT a)
12 const { return sign(a.x - x) == 0 && sign(a.y - y) ==
13 0; } bool operator != (PT a) const { return !(*this ==
14 a); } bool operator < (PT a) const { return sign(a.x
15 - x) == 0 ? y < a.y : x < a.x; } bool operator > (PT a)
16 const { return sign(a.x - x) == 0 ? y > a.y : x > a.
17 x; } double norm() { return sqrt(x * x + y * y); }
18 double norm2() { return x * x + y * y; } PT perp() {
19 return PT(-y, x); } double arg() { return atan2(y, x);
20 } PT truncate(double r) { // returns a vector with
21 norm r and having same direction double k = norm(); if
22 (!sign(k)) return *this; r /= k; return PT(x * r, y *
23 r); } };
24 istream &operator >> (istream &in, PT &p) { return in >> p.
25 x >> p.y; }
26 ostream &operator << (ostream &out, PT &p) { return out <<
27 "(" << p.x << ", " << p.y << ")"; }
28 inline double dot(PT a, PT b) { return a.x * b.x + a.y * b.
29 y; }
30 inline double dist2(PT a, PT b) { return dot(a - b, a - b);
31 }
32 inline double dist(PT a, PT b) { return sqrt(dot(a - b, a -
33 b)); }
34 inline double cross(PT a, PT b) { return a.x * b.y - a.y *
35 b.x; }
36 inline double cross2(PT a, PT b, PT c) { return cross(b - a
37 , c - a); }
38 inline int orientation(PT a, PT b, PT c) { return sign(
39 cross(b - a, c - a)); }
40 PT perp(PT a) { return PT(-a.y, a.x); }
41 PT rotateccw90(PT a) { return PT(-a.y, a.x); }
42 PT rotatecw90(PT a) { return PT(a.y, -a.x); }
43 PT rotateccw(PT a, double t) { return PT(a.x * cos(t) - a.y
44 * sin(t), a.x * sin(t) + a.y * cos(t)); }
45 PT rotatecw(PT a, double t) { return PT(a.x * cos(t) + a.y
46 * sin(t), -a.x * sin(t) + a.y * cos(t)); }
47 double SQ(double x) { return x * x; }
48 double rad_to_deg(double r) { return (r * 180.0 / PI); }
49 double deg_to_rad(double d) { return (d * PI / 180.0); }

```

```

18 double get_angle(PT a, PT b) {
19     double costheta = dot(a, b) / a.norm() / b.norm();
20     return acos(max((double)-1.0, min((double)1.0, costheta)
21 ));
22 }
23 bool is_point_in_angle(PT b, PT a, PT c, PT p) { // does
24     point p lie in angle <bac
25     assert(orientation(a, b, c) != 0);
26     if (orientation(a, c, b) < 0) swap(b, c);
27     return orientation(a, c, p) >= 0 && orientation(a, b, p)
28     <= 0;
29 }
30 bool half(PT p) {
31     return p.y > 0.0 || (p.y == 0.0 && p.x < 0.0);
32 }
33 void polar_sort(vector<PT> &v) { // sort points in
34     counterclockwise
35     sort(v.begin(), v.end(), [(PT a, PT b) {
36         return make_tuple(half(a), 0.0, a.norm2()) <
37         make_tuple(half(b), cross(a, b), b.norm2());
38     }]);
39 }
40 void polar_sort(vector<PT> &v, PT o) { // sort points in
41     counterclockwise with respect to point o
42     sort(v.begin(), v.end(), [(PT a, PT b) {
43         return make_tuple(half(a - o), 0.0, (a - o).norm2())
44         < make_tuple(half(b - o), cross(a - o, b - o), (
45         b - o).norm2());
46     }]);
47 }
48 struct line {
49     PT a, b; // goes through points a and b
50     PT v; double c; //line form: direction vec [cross] (x, y
51     ) = c
52     line() {}
53     //direction vector v and offset c
54     line(PT v, double c) : v(v), c(c) {
55         auto p = get_points();
56         a = p.first; b = p.second;
57     }
58     // equation ax + by + c = 0
59     line(double _a, double _b, double _c) : v({_b, -_a}), c(-
60     _c) {
61         auto p = get_points();
62         a = p.first; b = p.second;
63     }
64     // goes through points p and q
65     line(PT p, PT q) : v(q - p), c(cross(v, p)), a(p), b(q)
66     {}
67     pair<PT, PT> get_points() { //extract any two points
68         from this line
69         PT p, q; double a = -v.y, b = v.x; // ax + by = c
70         if (sign(a) == 0) {
71             p = PT(0, c / b); q = PT(1, c / b);
72         }
73         else if (sign(b) == 0) {
74             p = PT(c / a, 0); q = PT(c / a, 1);
75         }
76         else {
77             p = PT(0, c / b); q = PT(1, (c - a) / b);
78         }
79         return {p, q};
80     }
81     // ax + by + c = 0
82     array<double, 3> get_abc() {
83         double a = -v.y, b = v.x; return {a, b, -c};
84     }
85     // 1 if on the left, -1 if on the right, 0 if on the
86     line
87     int side(PT p) { return sign(cross(v, p) - c); }
88     // line that is perpendicular to this and goes through
89     point p
90     line perpendicular_through(PT p) { return {p, p + perp(v)
91     }); }
92     // translate the line by vector t i.e. shifting it by
93     vector t
94     line translate(PT t) { return {v, c + cross(v, t)}; }
95     // compare two points by their orthogonal projection on
96     this line
97     // a projection point comes before another if it comes
98     first according to vector v
99     bool cmp_by_projection(PT p, PT q) { return dot(v, p) <
100     dot(v, q); }
101     line shift_left(double d) {
102         PT z = v.perp().truncate(d);
103         return line(a + z, b + z);
104     }
105     // find a point from a through b with distance d
106     PT point_along_line(PT a, PT b, double d) {
107         assert(a != b);
108         return a + ((b - a) / (b - a).norm()) * d;
109     }
110     // projection point c onto line through a and b assuming a
111     != b
112     PT project_from_point_to_line(PT a, PT b, PT c) {
113         return a + (b - a) * dot(c - a, b - a) / (b - a).norm2()
114         ;
115     }
116     // reflection point c onto line through a and b assuming a
117     != b
118     PT reflection_from_point_to_line(PT a, PT b, PT c) {
119         PT p = project_from_point_to_line(a, b, c);
120         return p + p - c;
121     }
122     // minimum distance from point c to line through a and b
123     double dist_from_point_to_line(PT a, PT b, PT c) {
124         return fabs(cross(b - a, c - a) / (b - a).norm());
125     }
126     // returns true if point p is on line segment ab
127     bool is_point_on_seg(PT a, PT b, PT p) {
128         if (fabs(cross(p - b, a - b)) < eps) {
129             if (p.x < min(a.x, b.x) - eps || p.x > max(a.x, b.x)
130             + eps) return false;
131             if (p.y < min(a.y, b.y) - eps || p.y > max(a.y, b.y)
132             + eps) return false;
133             return true;
134         }
135         return false;
136     }
137     // minimum distance point from point c to segment ab that
138     lies on segment ab
139     PT project_from_point_to_seg(PT a, PT b, PT c) {
140         double r = dist2(a, b);
141         if (sign(r) == 0) return a;
142         r = dot(c - a, b - a) / r;
143         if (r < 0) return a;
144         if (r > 1) return b;
145         return a + (b - a) * r;
146     }
147     // minimum distance from point c to segment ab
148     double dist_from_point_to_seg(PT a, PT b, PT c) {
149         return dist(c, project_from_point_to_seg(a, b, c));
150     }
151     // 0 if not parallel, 1 if parallel, 2 if collinear
152     int is_parallel(PT a, PT b, PT c, PT d) {
153         double k = fabs(cross(b - a, d - c));
154         if (k < eps) {
155             if (fabs(cross(a - b, a - c)) < eps && fabs(cross(c -
156             d, c - a)) < eps) return 2;
157             else return 1;
158         }
159         else return 0;
160     }
161     // check if two lines are same
162     bool are_lines_same(PT a, PT b, PT c, PT d) {
163         if (fabs(cross(a - c, c - d)) < eps && fabs(cross(b - c,
164         c - d)) < eps) return true;
165         return false;
166     }
167     // bisector vector of <abc
168     PT angle_bisector(PT &a, PT &b, PT &c) {
169         PT p = a - b, q = c - b;
170         return p + q * sqrt(dot(p, p) / dot(q, q));
171     }
172     // 1 if point is ccw to the line, 2 if point is cw to the
173     line, 3 if point is on the line
174     int point_line_relation(PT a, PT b, PT p) {
175         int c = sign(cross(p - a, b - a));
176         if (c < 0) return 1;
177         if (c > 0) return 2;
178         return 3;
179     }
180     // intersection point between ab and cd assuming unique
181     intersection exists
182     bool line_line_intersection(PT a, PT b, PT c, PT d, PT &ans
183     ) {
184         double a1 = a.y - b.y, b1 = b.x - a.x, c1 = cross(a, b);
185         double a2 = c.y - d.y, b2 = d.x - c.x, c2 = cross(c, d);
186         double det = a1 * b2 - a2 * b1;
187         if (det == 0) return 0;
188         ans = PT((b1 * c2 - b2 * c1) / det, (c1 * a2 - a1 * c2)
189         / det);
190         return 1;
191     }
192     // intersection point between segment ab and segment cd
193     assuming unique intersection exists
194     bool seg_seg_intersection(PT a, PT b, PT c, PT d, PT &ans)
195     {
196         double oa = cross2(c, d, a), ob = cross2(c, d, b);
197         double oc = cross2(a, b, c), od = cross2(a, b, d);
198         if (oa * ob < 0 && oc * od < 0) {
199             ans = (a * ob - b * oa) / (ob - oa);
200             return 1;
201         }
202         else return 0;
203     }
204     // intersection point between segment ab and segment cd
205     assuming unique intersection may not exists
206     // se.size()==0 means no intersection
207     // se.size()==1 means one intersection
208     // se.size()==2 means range intersection
209     set<PT> seg_seg_intersection_inside(PT a, PT b, PT c, PT d)
210     {
211         PT ans;

```

```

178 if (seg_seg_intersection(a, b, c, d, ans)) return {ans};
179 set<PT> se;
180 if (is_point_on_seg(c, d, a)) se.insert(a);
181 if (is_point_on_seg(c, d, b)) se.insert(b);
182 if (is_point_on_seg(a, b, c)) se.insert(c);
183 if (is_point_on_seg(a, b, d)) se.insert(d);
184 return se;
185 }
186 // intersection between segment ab and line cd
187 // 0 if do not intersect, 1 if proper intersect, 2 if
    segment intersect
188 int seg_line_relation(PT a, PT b, PT c, PT d) {
189     double p = cross2(c, d, a);
190     double q = cross2(c, d, b);
191     if (sign(p) == 0 && sign(q) == 0) return 2;
192     else if (p * q < 0) return 1;
193     else return 0;
194 }
195 // intersection between segment ab and line cd assuming
    unique intersection exists
196 bool seg_line_intersection(PT a, PT b, PT c, PT d, PT &ans)
    {
197     bool k = seg_line_relation(a, b, c, d);
198     assert(k != 2);
199     if (k) line_line_intersection(a, b, c, d, ans);
200     return k;
201 }
202 // minimum distance from segment ab to segment cd
203 double dist_from_seg_to_seg(PT a, PT b, PT c, PT d) {
204     PT dummy;
205     if (seg_seg_intersection(a, b, c, d, dummy)) return 0.0;
206     else return min({dist_from_point_to_seg(a, b, c),
207                     dist_from_point_to_seg(a, b, d),
208                     dist_from_point_to_seg(c, d, a),
209                     dist_from_point_to_seg(c, d, b)});
210 }
211 // minimum distance from point c to ray (starting point a
    and direction vector b)
212 double dist_from_point_to_ray(PT a, PT b, PT c) {
213     b = a + b;
214     double r = dot(c - a, b - a);
215     if (r < 0.0) return dist(c, a);
216     return dist_from_point_to_line(a, b, c);
217 }
218 // starting point as and direction vector ad
219 bool ray_ray_intersection(PT as, PT ad, PT bs, PT bd) {
220     double dx = bs.x - as.x, dy = bs.y - as.y;
221     double det = bd.x * ad.y - bd.y * ad.x;
222     if (fabs(det) < eps) return 0;
223     double u = (dy * bd.x - dx * bd.y) / det;
224     double v = (dy * ad.x - dx * ad.y) / det;
225     if (sign(u) >= 0 && sign(v) >= 0) return 1;
226     else return 0;
227 }
228 double ray_ray_distance(PT as, PT ad, PT bs, PT bd) {
229     if (ray_ray_intersection(as, ad, bs, bd)) return 0.0;
230     double ans = dist_from_point_to_ray(as, ad, bs);
231     ans = min(ans, dist_from_point_to_ray(bs, bd, as));
232     return ans;
233 }
234 struct circle {
235     PT p; double r;
236     circle() {}
237     circle(PT _p, double _r): p(_p), r(_r) {};
238     // center (x, y) and radius r
239
240     circle(double x, double y, double _r): p(PT(x, y)), r(_r)
        {}
241     // circumcircle of a triangle
242     // the three points must be unique
243     circle(PT a, PT b, PT c) {
244         b = (a + b) * 0.5;
245         c = (a + c) * 0.5;
246         line_line_intersection(b, b + rotatecw90(a - b), c, c
            + rotatecw90(a - c), p);
247         r = dist(a, p);
248     }
249     // inscribed circle of a triangle
250     // pass a bool just to differentiate from circumcircle
251     circle(PT a, PT b, PT c, bool t) {
252         line u, v;
253         double m = atan2(b.y - a.y, b.x - a.x), n = atan2(c.y
            - a.y, c.x - a.x);
254         u.a = a;
255         u.b = u.a + (PT(cos((n + m)/2.0), sin((n + m)/2.0)));
256         v.a = b;
257         m = atan2(a.y - b.y, a.x - b.x), n = atan2(c.y - b.y,
            c.x - b.x);
258         v.b = v.a + (PT(cos((n + m)/2.0), sin((n + m)/2.0)));
259         line_line_intersection(u.a, u.b, v.a, v.b, p);
260         r = dist_from_point_to_seg(a, b, p);
261     }
262     bool operator == (circle v) { return p == v.p && sign(r
        - v.r) == 0; }
263     double area() { return PI * r * r; }
264     double circumference() { return 2.0 * PI * r; }
265 };
266 //0 if outside, 1 if on circumference, 2 if inside circle
267 int circle_point_relation(PT p, double r, PT b) {
268     double d = dist(p, b);
269     if (sign(d - r) < 0) return 2;
270     if (sign(d - r) == 0) return 1;
271     return 0;
272 }
273 // 0 if outside, 1 if on circumference, 2 if inside circle
274 int circle_line_relation(PT p, double r, PT a, PT b) {
275     double d = dist_from_point_to_line(a, b, p);
276     if (sign(d - r) < 0) return 2;
277     if (sign(d - r) == 0) return 1;
278     return 0;
279 }
280 //compute intersection of line through points a and b with
    //circle centered at c with radius r > 0
281 vector<PT> circle_line_intersection(PT c, double r, PT a,
    PT b) {
282     vector<PT> ret;
283     b = b - a; a = a - c;
284     double A = dot(b, b), B = dot(a, b);
285     double C = dot(a, a) - r * r, D = B * B - A * C;
286     if (D < -eps) return ret;
287     ret.push_back(c + a + b * (-B + sqrt(D + eps)) / A);
288     if (D > eps) ret.push_back(c + a + b * (-B - sqrt(D)) /
        A);
289     return ret;
290 }
291 //5 - outside and do not intersect
292 //4 - intersect outside in one point
293 //3 - intersect in 2 points
294 //2 - intersect inside in one point
295 //1 - inside and do not intersect
296
297 int circle_circle_relation(PT a, double r, PT b, double R)
    {
298     double d = dist(a, b);
299     if (sign(d - r - R) > 0) return 5;
300     if (sign(d - r - R) == 0) return 4;
301     double l = fabs(r - R);
302     if (sign(d - r - R) < 0 && sign(d - l) > 0) return 3;
303     if (sign(d - l) == 0) return 2;
304     if (sign(d - l) < 0) return 1;
305     assert(0); return -1;
306 }
307 vector<PT> circle_circle_intersection(PT a, double r, PT b,
    double R) {
308     if (a == b && sign(r - R) == 0) return {PT(1e18, 1e18)};
309     vector<PT> ret;
310     double d = sqrt(dist2(a, b));
311     if (d > r + R || d + min(r, R) < max(r, R)) return ret;
312     double x = (d * d - R * R + r * r) / (2 * d);
313     double y = sqrt(r * r - x * x);
314     PT v = (b - a) / d;
315     ret.push_back(a + v * x + rotateccw90(v) * y);
316     if (y > 0) ret.push_back(a + v * x - rotateccw90(v) * y);
317     return ret;
318 }
319 // returns two circle c1, c2 through points a, b and of
    radius r
320 // 0 if there is no such circle, 1 if one circle, 2 if two
    circle
321 int get_circle(PT a, PT b, double r, circle &c1, circle &c2)
    {
322     vector<PT> v = circle_circle_intersection(a, r, b, r);
323     int t = v.size();
324     if (!t) return 0;
325     c1.p = v[0], c1.r = r;
326     if (t == 2) c2.p = v[1], c2.r = r;
327     return t;
328 }
329 // returns two circle c1, c2 which is tangent to line u,
    goes through
330 // point q and has radius r1; 0 for no circle, 1 if c1 = c2
    , 2 if c1 != c2
331 int get_circle(line u, PT q, double r1, circle &c1, circle
    &c2) {
332     double d = dist_from_point_to_line(u.a, u.b, q);
333     if (sign(d - r1 * 2.0) > 0) return 0;
334     if (sign(d) == 0) {
335         cout << u.v.x << ' ' << u.v.y << '\n';
336         c1.p = q + rotateccw90(u.v).truncate(r1);
337         c2.p = q + rotatecw90(u.v).truncate(r1);
338         c1.r = c2.r = r1;
339         return 2;
340     }
341     line u1 = line(u.a + rotateccw90(u.v).truncate(r1), u.b
        + rotateccw90(u.v).truncate(r1));
342     line u2 = line(u.a + rotatecw90(u.v).truncate(r1), u.b +
        rotatecw90(u.v).truncate(r1));
343     circle cc = circle(q, r1);
344     PT p1, p2; vector<PT> v;
345     v = circle_line_intersection(q, r1, u1.a, u1.b);
346     if (!v.size()) v = circle_line_intersection(q, r1, u2.a,
        u2.b);
347     v.push_back(v[0]);
348     p1 = v[0], p2 = v[1];
349     c1 = circle(p1, r1);
350 }

```

```

347     if (p1 == p2) {
348         c2 = c1;
349         return 1;
350     }
351     c2 = circle(p2, r1);
352     return 2;
353 }
354
355 // returns area of intersection between two circles
356 double circle_circle_area(PT a, double r1, PT b, double r2)
357 {
358     double d = (a - b).norm();
359     if (r1 + r2 < d + eps) return 0;
360     if (r1 + d < r2 + eps) return PI * r1 * r1;
361     if (r2 + d < r1 + eps) return PI * r2 * r2;
362     double theta_1 = acos((r1 * r1 + d * d - r2 * r2) / (2 *
363         r1 * d)),
364         theta_2 = acos((r2 * r2 + d * d - r1 * r1) / (2 * r2 * d));
365     return r1 * r1 * (theta_1 - sin(2 * theta_1) / 2.) + r2 *
366         r2 * (theta_2 - sin(2 * theta_2) / 2.);
367 }
368 // tangent lines from point q to the circle
369 int tangent_lines_from_point(PT p, double r, PT q, line &u,
370     line &v) {
371     int x = sign(dist2(p, q) - r * r);
372     if (x < 0) return 0; // point in circle
373     if (x == 0) { // point on circle
374         u = line(q, q + rotateccw90(q - p));
375         v = u;
376         return 1;
377     }
378     double d = dist(p, q);
379     double l = r * r / d;
380     double h = sqrt(r * r - l * l);
381     u = line(q, p + ((q - p).truncate(l) + (rotateccw90(q -
382         p).truncate(h))));
383     v = line(q, p + ((q - p).truncate(l) + (rotatecw90(q - p
384         ).truncate(h))));
385     return 2;
386 }
387 // returns outer tangents line of two circles
388 // if inner == 1 it returns inner tangent lines
389 int tangents_lines_from_circle(PT c1, double r1, PT c2,
390     double r2, bool inner, line &u, line &v) {
391     if (inner) r2 = -r2;
392     PT d = c2 - c1;
393     double dr = r1 - r2, d2 = d.norm2(), h2 = d2 - dr * dr;
394     if (d2 == 0 || h2 < 0) {
395         assert(h2 != 0);
396         return 0;
397     }
398     vector<pair<PT, PT>>out;
399     for (int tmp: {-1, 1}) {
400         PT v = (d * dr + rotateccw90(d) * sqrt(h2) * tmp) /
401             d2;
402         out.push_back({c1 + v * r1, c2 + v * r2});
403     }
404     u = line(out[0].first, out[0].second);
405     if (out.size() == 2) v = line(out[1].first, out[1].
406         second);
407     return 1 + (h2 > 0);
408 }
409
410 double area_of_triangle(PT a, PT b, PT c) {
411     return fabs(cross(b - a, c - a) * 0.5);
412 }
413 // -1 if strictly inside, 0 if on the polygon, 1 if
414 // strictly outside
415 int is_point_in_triangle(PT a, PT b, PT c, PT p) {
416     if (sign(cross(b - a, c - a)) < 0) swap(b, c);
417     int c1 = sign(cross(b - a, p - a));
418     int c2 = sign(cross(c - b, p - b));
419     int c3 = sign(cross(a - c, p - c));
420     if (c1 < 0 || c2 < 0 || c3 < 0) return 1;
421     if (c1 + c2 + c3 != 3) return 0;
422     return -1;
423 }
424 double perimeter(vector<PT> &p) {
425     double ans = 0; int n = p.size();
426     for (int i = 0; i < n; i++) ans += dist(p[i], p[(i + 1)
427         % n]);
428     return ans;
429 }
430 double area(vector<PT> &p) {
431     double ans = 0; int n = p.size();
432     for (int i = 0; i < n; i++) ans += cross(p[i], p[(i + 1)
433         % n]);
434     return fabs(ans) * 0.5;
435 }
436 // centroid of a (possibly non-convex) polygon,
437 // assuming that the coordinates are listed in a clockwise
438 // or
439 // counterclockwise fashion. Note that the centroid is
440 // often known as
441 // the "center of gravity" or "center of mass".
442 PT centroid(vector<PT> &p) {
443     int n = p.size(); PT c(0, 0);
444     double sum = 0;
445     for (int i = 0; i < n; i++) sum += cross(p[i], p[(i + 1)
446         % n]);
447     double scale = 3.0 * sum;
448     for (int i = 0; i < n; i++) {
449         int j = (i + 1) % n;
450         c = c + (p[i] + p[j]) * cross(p[i], p[j]);
451     }
452     return c / scale;
453 }
454 // 0 if cw, 1 if ccw
455 bool get_direction(vector<PT> &p) {
456     double ans = 0; int n = p.size();
457     for (int i = 0; i < n; i++) ans += cross(p[i], p[(i + 1)
458         % n]);
459     if (sign(ans) > 0) return 1;
460     return 0;
461 }
462 // it returns a point such that the sum of distances
463 // from that point to all points in p is minimum
464 // O(n log^2 MX)
465 PT geometric_median(vector<PT> p) {
466     auto tot_dist = [&](PT z) {
467         double res = 0;
468         for (int i = 0; i < p.size(); i++) res += dist(p[i], z
469             );
470         return res;
471     };
472     auto findY = [&](double x) {
473         double y1 = -1e5, yr = 1e5;
474         for (int i = 0; i < 60; i++) {
475             double ym1 = y1 + (yr - y1) / 3;
476             double ym2 = yr - (yr - y1) / 3;
477             double d1 = tot_dist(PT(x, ym1));
478             double d2 = tot_dist(PT(x, ym2));
479             if (d1 < d2) yr = ym2;
480             else y1 = ym1;
481         }
482         return pair<double, double> (y1, tot_dist(PT(x, y1)));
483     };
484     double x1 = -1e5, xr = 1e5;
485     for (int i = 0; i < 60; i++) {
486         double xm1 = x1 + (xr - x1) / 3;
487         double xm2 = xr - (xr - x1) / 3;
488         double y1, d1, y2, d2;
489         auto z = findY(xm1); y1 = z.first; d1 = z.second;
490         z = findY(xm2); y2 = z.first; d2 = z.second;
491         if (d1 < d2) xr = xm2;
492         else x1 = xm1;
493     }
494     return {x1, findY(x1).first};
495 }
496 // -1 if strictly inside, 0 if on the polygon, 1 if
497 // strictly outside
498 // it must be strictly convex, otherwise make it strictly
499 // convex first
500 int is_point_in_convex(vector<PT> &p, const PT& x) { // O(
501     log n)
502     int n = p.size(); assert(n >= 3);
503     int a = orientation(p[0], p[1], x), b = orientation(p
504         [0], p[n - 1], x);
505     if (a < 0 || b > 0) return 1;
506     int l = 1, r = n - 1;
507     while (l + 1 < r) {
508         int mid = l + r >> 1;
509         if (orientation(p[0], p[mid], x) >= 0) l = mid;
510         else r = mid;
511     }
512     int k = orientation(p[1], p[r], x);
513     if (k <= 0) return -k;
514     if (l == 1 && a == 0) return 0;
515     if (r == n - 1 && b == 0) return 0;
516     return -1;
517 }
518 bool is_point_on_polygon(vector<PT> &p, const PT& z) {
519     int n = p.size();
520     for (int i = 0; i < n; i++) {
521         if (is_point_on_seg(p[i], p[(i + 1) % n], z)) return
522             1;
523     }
524     return 0;
525 }
526 // returns 1e9 if the point is on the polygon
527 int winding_number(vector<PT> &p, const PT& z) { // O(n)
528     if (is_point_on_polygon(p, z)) return 1e9;
529     int n = p.size(), ans = 0;
530     for (int i = 0; i < n; i++) {
531         int j = (i + 1) % n;
532         bool below = p[i].y < z.y;
533         if (below != (p[j].y < z.y)) {
534             auto orient = orientation(z, p[j], p[i]);
535             if (orient == 0) return 0;
536             if (below == (orient > 0)) ans += below ? 1 : -1;
537         }
538     }
539     return ans;
540 }

```

```

519 }
520 // -1 if strictly inside, 0 if on the polygon, 1 if
521 // strictly outside
522 int is_point_in_polygon(vector<PT> &p, const PT& z) { // O(n)
523     int k = winding_number(p, z);
524     return k == 1e9 ? 0 : k == 0 ? 1 : -1;
525 }
526 double diameter(vector<PT> &p) {
527     int n = (int)p.size();
528     if (n == 1) return 0;
529     if (n == 2) return dist(p[0], p[1]);
530     double ans = 0;
531     int i = 0, j = 1;
532     while (i < n) {
533         while (cross(p[(i + 1) % n] - p[i], p[(j + 1) % n] -
534             p[j]) >= 0) {
535             ans = max(ans, dist2(p[i], p[j]));
536             j = (j + 1) % n;
537         }
538         ans = max(ans, dist2(p[i], p[j]));
539         i++;
540     }
541     return sqrt(ans);
542 }
543 // minimum distance between two parallel lines (non
544 // necessarily axis parallel)
545 // such that the polygon can be put between the lines
546 double width(vector<PT> &p) {
547     int n = (int)p.size();
548     if (n <= 2) return 0;
549     double ans = inf;
550     int i = 0, j = 1;
551     while (i < n) {
552         while (cross(p[(i + 1) % n] - p[i], p[(j + 1) % n] -
553             p[j]) >= 0) j = (j + 1) % n;
554         ans = min(ans, dist_from_point_to_line(p[i], p[(i +
555             1) % n], p[j]));
556         i++;
557     }
558     return ans;
559 }
560 // minimum perimeter
561 double minimum_enclosing_rectangle(vector<PT> &p) {
562     int n = p.size();
563     if (n <= 2) return perimeter(p);
564     int mndot = 0; double tmp = dot(p[1] - p[0], p[0]);
565     for (int i = 1; i < n; i++) {
566         if (dot(p[1] - p[0], p[i]) <= tmp) {
567             tmp = dot(p[1] - p[0], p[i]);
568             mndot = i;
569         }
570     }
571     double ans = inf;
572     int i = 0, j = 1, mxdot = 1;
573     while (i < n) {
574         PT cur = p[(i + 1) % n] - p[i];
575         while (cross(cur, p[(j + 1) % n] - p[j]) >= 0) j = (j
576             + 1) % n;
577         while (dot(p[(mxdot + 1) % n], cur) >= dot(p[mxdot],
578             cur)) mxdot = (mxdot + 1) % n;
579         while (dot(p[(mndot + 1) % n], cur) <= dot(p[mndot],
580             cur)) mndot = (mndot + 1) % n;
581         ans = min(ans, 2.0 * ((dot(p[mxdot], cur) / cur.norm
582             ()) - dot(p[mndot], cur) / cur.norm()) +
583             dist_from_point_to_line(p[i], p[(i + 1) % n], p[
584                 j]));
585         i++;
586     }
587     return ans;
588 }
589 // given n points, find the minimum enclosing circle of the
590 // points
591 // call convex_hull() before this for faster solution
592 // expected O(n)
593 circle minimum_enclosing_circle(vector<PT> &p) {
594     random_shuffle(p.begin(), p.end());
595     int n = p.size();
596     circle c(p[0], 0);
597     for (int i = 1; i < n; i++) {
598         if (sign(dist(c.p, p[i]) - c.r) > 0) {
599             c = circle(p[i], 0);
600             for (int j = 0; j < i; j++) {
601                 if (sign(dist(c.p, p[j]) - c.r) > 0) {
602                     c = circle((p[i] + p[j]) / 2, dist(p[i], p[j]
603                         ) / 2);
604                     for (int k = 0; k < j; k++) {
605                         if (sign(dist(c.p, p[k]) - c.r) > 0) {
606                             c = circle(p[i], p[j], p[k]);
607                         }
608                     }
609                 }
610             }
611         }
612     }
613     return c;
614 }
615 pair<PT, int> point_poly_tangent(vector<PT> &p, PT Q, int
616     dir, int l, int r) {
617     while (r - l > 1) {
618         int mid = (l + r) >> 1;
619         bool pvs = orientation(Q, p[mid], p[mid - 1]) != -dir
620             ;
621         bool nxt = orientation(Q, p[mid], p[mid + 1]) != -dir
622             ;
623         if (pvs && nxt) return {p[mid], mid};
624         if (!(pvs || nxt)) {
625             auto p1 = point_poly_tangent(p, Q, dir, mid + 1, r
626                 );
627             auto p2 = point_poly_tangent(p, Q, dir, l, mid -
628                 1);
629             return orientation(Q, p1.first, p2.first) == dir ?
630                 p1 : p2;
631         }
632         if (!pvs) {
633             if (orientation(Q, p[mid], p[l]) == dir) r = mid -
634                 1;
635             else if (orientation(Q, p[l], p[r]) == dir) r =
636                 mid - 1;
637             else l = mid + 1;
638         }
639         if (!nxt) {
640             if (orientation(Q, p[mid], p[l]) == dir) l = mid +
641                 1;
642             else if (orientation(Q, p[l], p[r]) == dir) r =
643                 mid - 1;
644         }
645     }
646     return {p[l], l};
647 }
648 // (ccw, cw) tangents from a point that is outside this
649 // convex polygon
650 // returns indexes of the points
651 // ccw means the tangent from Q to that point is in the
652 // same direction as the polygon ccw direction
653 pair<int, int> tangents_from_point_to_polygon(vector<PT> &p
654     , PT Q) {
655     int ccw = point_poly_tangent(p, Q, 1, 0, (int)p.size() -
656         1).second;
657     int cw = point_poly_tangent(p, Q, -1, 0, (int)p.size() -
658         1).second;
659     return make_pair(ccw, cw);
660 }
661 // minimum distance from a point to a convex polygon
662 // it assumes point lie strictly outside the polygon
663 double dist_from_point_to_polygon(vector<PT> &p, PT z) {
664     double ans = inf;
665     int n = p.size();
666     if (n <= 3) {
667         for (int i = 0; i < n; i++) ans = min(ans,
668             dist_from_point_to_seg(p[i], p[(i + 1) % n], z));
669     }
670     return ans;
671 }
672 auto [r, l] = tangents_from_point_to_polygon(p, z);
673 if (l > r) r += n;
674 while (l < r) {
675     int mid = (l + r) >> 1;
676     double left = dist2(p[mid % n], z), right = dist2(p[(
677         mid + 1) % n], z);
678     ans = min({ans, left, right});
679     if (left < right) r = mid;
680     else l = mid + 1;
681 }
682 ans = sqrt(ans);
683 ans = min(ans, dist_from_point_to_seg(p[l % n], p[(l +
684     1) % n], z));
685 ans = min(ans, dist_from_point_to_seg(p[l % n], p[(l - 1
686     + n) % n], z));
687 return ans;
688 }
689 // minimum distance from convex polygon p to line ab
690 // returns 0 is it intersects with the polygon
691 // top - upper right vertex
692 double dist_from_polygon_to_line(vector<PT> &p, PT a, PT b,
693     int top) { //O(log n)
694     PT orth = (b - a).perp();
695     if (orientation(a, b, p[0]) > 0) orth = (a - b).perp();
696     int id = extreme_vertex(p, orth, top);
697     if (dot(p[id] - a, orth) > 0) return 0.0; //if orth and a
698     // are in the same half of the line, then poly and
699     // line intersects
700     return dist_from_point_to_line(a, b, p[id]); //does not
701     // intersect
702 }

```

```

673 // minimum distance from a convex polygon to another convex
674 // polygon
675 // the polygon doesnot overlap or touch
676 // tested in https://toph.co/p/the-wall
677 double dist_from_polygon_to_polygon(vector<PT> &p1, vector<
678 PT> &p2) { // O(n log n)
679     double ans = inf;
680     for (int i = 0; i < p1.size(); i++) {
681         ans = min(ans, dist_from_point_to_polygon(p2, p1[i]))
682     };
683     for (int i = 0; i < p2.size(); i++) {
684         ans = min(ans, dist_from_point_to_polygon(p1, p2[i]))
685     };
686     return ans;
687 }
688 // maximum distance from a convex polygon to another convex
689 // polygon
690 double maximum_dist_from_polygon_to_polygon(vector<PT> &u,
691 vector<PT> &v) { //O(n)
692     int n = (int)u.size(), m = (int)v.size();
693     double ans = 0;
694     if (n < 3 || m < 3) {
695         for (int i = 0; i < n; i++) {
696             for (int j = 0; j < m; j++) ans = max(ans, dist2(u
697 [i], v[j]));
698         }
699         return sqrt(ans);
700     }
701     if (u[0].x > v[0].x) swap(n, m), swap(u, v);
702     int i = 0, j = 0, step = n + m + 10;
703     while (j + 1 < m && v[j].x < v[j + 1].x) j++;
704     while (step--) {
705         if (cross(u[(i + 1) % n] - u[i], v[(j + 1) % m] - v[j])
706             >= 0) j = (j + 1) % m;
707         else i = (i + 1) % n;
708         ans = max(ans, dist2(u[i], v[j]));
709     }
710     return sqrt(ans);
711 }
712 // rotate the polygon such that the (bottom, left)-most
713 // point is at the first position
714 void reorder_polygon(vector<PT> &p) {
715     int pos = 0;
716     for (int i = 1; i < p.size(); i++) {
717         if (p[i].y < p[pos].y || (sign(p[i].y - p[pos].y) == 0
718             && p[i].x < p[pos].x)) pos = i;
719     }
720     rotate(p.begin(), p.begin() + pos, p.end());
721 }
722 // returns the area of the intersection of the circle with
723 // center c and radius r
724 // and the triangle formed by the points c, a, b
725 double _triangle_circle_intersection(PT c, double r, PT a,
726 PT b) {
727     double sd1 = dist2(c, a), sd2 = dist2(c, b);
728     if (sd1 > sd2) swap(a, b), swap(sd1, sd2);
729     double sd = dist2(a, b);
730     double d1 = sqrt1(sd1), d2 = sqrt1(sd2), d = sqrt(sd);
731     double x = abs(sd2 - sd - sd1) / (2 * d);
732     double h = sqrt1(sd1 - x * x);
733     if (r >= d2) return h * d / 2;
734     double area = 0;

```

```

726 if (sd + sd1 < sd2) {
727     if (r < d1) area = r * r * (acos(h / d2) - acos(h / d1
728 )) / 2;
729     else {
730         area = r * r * (acos(h / d2) - acos(h / r)) / 2;
731         double y = sqrt1(r * r - h * h);
732         area += h * (y - x) / 2;
733     }
734 }
735 else {
736     if (r < h) area = r * r * (acos(h / d2) + acos(h / d1
737 )) / 2;
738     else {
739         area += r * r * (acos(h / d2) - acos(h / r)) / 2;
740         double y = sqrt1(r * r - h * h);
741         area += h * y / 2;
742         if (r < d1) {
743             area += r * r * (acos(h / d1) - acos(h / r)) /
744 2;
745             area += h * y / 2;
746         }
747         else area += h * x / 2;
748     }
749 }
750 return area;
751 }
752 // intersection between a simple polygon and a circle
753 double polygon_circle_intersection(vector<PT> &v, PT p,
754 double r) {
755     int n = v.size();
756     double ans = 0.00;
757     PT org = {0, 0};
758     for (int i = 0; i < n; i++) {
759         int x = orientation(p, v[i], v[(i + 1) % n]);
760         if (x == 0) continue;
761         double area = _triangle_circle_intersection(org, r, v
762 [i] - p, v[(i + 1) % n] - p);
763         if (x < 0) ans -= area;
764         else ans += area;
765     }
766     return abs(ans);
767 }
768 // find a circle of radius r that contains as many points
769 // as possible
770 // O(n^2 log n);
771 double maximum_circle_cover(vector<PT> p, double r, circle
772 &c) {
773     int n = p.size();
774     int ans = 0;
775     int id = 0; double th = 0;
776     for (int i = 0; i < n; ++i) {
777         // maximum circle cover when the circle goes through
778         // this point
779         vector<pair<double, int>> events = {{-PI, +1}, {PI,
780 -1}};
781         for (int j = 0; j < n; ++j) {
782             if (j == i) continue;
783             double d = dist(p[i], p[j]);
784             if (d > r * 2) continue;
785             double dir = (p[j] - p[i]).arg();
786             double ang = acos(d / 2 / r);
787             double st = dir - ang, ed = dir + ang;
788             if (st > PI) st -= PI * 2;
789             if (st <= -PI) st += PI * 2;
790             if (ed > PI) ed -= PI * 2;

```

```

782 if (ed <= -PI) ed += PI * 2;
783 events.push_back({st - eps, +1}); // take care of
784 // precisions!
785 events.push_back({ed, -1});
786 if (st > ed) {
787     events.push_back({-PI, +1});
788     events.push_back({+PI, -1});
789 }
790 }
791 sort(events.begin(), events.end());
792 int cnt = 0;
793 for (auto &&e: events) {
794     cnt += e.second;
795     if (cnt > ans) {
796         ans = cnt;
797         id = i; th = e.first;
798     }
799 }
800 PT w = PT(p[id].x + r * cos(th), p[id].y + r * sin(th));
801 c = circle(w, r); //best_circle
802 return ans;
803 }
804 // radius of the maximum inscribed circle in a convex
805 // polygon
806 double maximum_inscribed_circle(vector<PT> p) {
807     int n = p.size();
808     if (n <= 2) return 0;
809     double l = 0, r = 20000;
810     while (r - l > eps) {
811         double mid = (l + r) * 0.5;
812         vector<HP> h;
813         const int L = 1e9;
814         h.push_back(HP(PT(-L, -L), PT(L, -L)));
815         h.push_back(HP(PT(L, -L), PT(L, L)));
816         h.push_back(HP(PT(L, L), PT(-L, L)));
817         h.push_back(HP(PT(-L, L), PT(-L, -L)));
818         for (int i = 0; i < n; i++) {
819             PT z = (p[(i + 1) % n] - p[i]).perp();
820             z = z.truncate(mid);
821             PT y = p[i] + z, q = p[(i + 1) % n] + z;
822             h.push_back(HP(p[i] + z, p[(i + 1) % n] + z));
823         }
824         vector<PT> nw = half_plane_intersection(h);
825         if (!nw.empty()) l = mid;
826         else r = mid;
827     }
828     return l;
829 }

```

## Graph

### Articulation Point Bridge

```

1 int n; // number of nodes
2 vector<vector<int>> adj; // adjacency list of graph
3 vector<bool> visited;
4 vector<int> tin, low;
5 int timer;
6 void dfs(int v, int p = -1) {
7     visited[v] = true;
8     tin[v] = low[v] = timer++;
9     int children=0;
10    for (int to : adj[v]) {

```

```

11     if (to == p) continue;
12     if (visited[to]) {
13         low[v] = min(low[v], tin[to]);
14     } else {
15         dfs(to, v);
16         low[v] = min(low[v], low[to]);
17         if (low[to] >= tin[v] && p != -1)
18             IS_CUTPOINT(v);
19         ++children;
20     }
21 }
22 if (p == -1 && children > 1)
23     IS_CUTPOINT(v);
24 }
25 void find_cutpoints() {
26     timer = 0;
27     visited.assign(n, false);
28     tin.assign(n, -1);
29     low.assign(n, -1);
30     for (int i = 0; i < n; ++i) {
31         if (!visited[i])
32             dfs(i);
33     }
34 }
35 // finding bridge.
36 void IS_BRIDGE(int v, int to); // some function to process
37 // the found bridge
38 int n; // number of nodes
39 vector<vector<int>> adj; // adjacency list of graph
40 vector<bool> visited;
41 vector<int> tin, low;
42 int timer;
43 void dfs(int v, int p = -1) {
44     visited[v] = true;
45     tin[v] = low[v] = timer++;
46     bool parent_skipped = false;
47     for (int to : adj[v]) {
48         if (to == p && !parent_skipped) {
49             parent_skipped = true;
50             continue;
51         }
52         if (visited[to]) {
53             low[v] = min(low[v], tin[to]);
54         } else {
55             dfs(to, v);
56             low[v] = min(low[v], low[to]);
57             if (low[to] > tin[v])
58                 IS_BRIDGE(v, to);
59         }
60     }
61 }
62 void find_bridges() {
63     timer = 0;
64     visited.assign(n, false);
65     tin.assign(n, -1);
66     low.assign(n, -1);
67     for (int i = 0; i < n; ++i) {
68         if (!visited[i])
69             dfs(i);
70     }

```

Bell

```

1 void solve()
2 {
3     vector<int> d(n, INF);
4     d[v] = 0;
5     vector<int> p(n, -1);
6     int x;
7     for (int i = 0; i < n; ++i) {
8         x = -1;
9         for (Edge e : edges)
10             if (d[e.a] < INF)
11                 if (d[e.b] > d[e.a] + e.cost) {
12                     d[e.b] = max(-INF, d[e.a] + e.cost);
13                     p[e.b] = e.a;
14                     x = e.b;
15                 }
16     }
17     if (x == -1)
18         cout << "No negative cycle from " << v;
19     else {
20         int y = x;
21         for (int i = 0; i < n; ++i)
22             y = p[y];
23         vector<int> path;
24         for (int cur = y;; cur = p[cur]) {
25             path.push_back(cur);
26             if (cur == y && path.size() > 1)
27                 break;
28         }
29         reverse(path.begin(), path.end());
30         cout << "Negative cycle: ";
31         for (int u : path)
32             cout << u << ' ';
33     }
34 }

```

## Block Cut Tree

```

1 #include<bits/stdc++.h>
2 using namespace std;
3
4 const int N = 4e5 + 9;
5
6 int T, low[N], dis[N], art[N], sz;
7 vector<int> g[N], bcc[N], st;
8 void dfs(int u, int pre = 0) {
9     low[u] = dis[u] = ++T;
10    st.push_back(u);
11    for (auto v : g[u]) {
12        if (!dis[v]) {
13            dfs(v, u);
14            low[u] = min(low[u], low[v]);
15            if (low[v] >= dis[u]) {
16                sz++;
17                int x;
18                do {
19                    x = st.back();
20                    st.pop_back();
21                    bcc[x].push_back(sz);
22                } while (x != v);
23                bcc[u].push_back(sz);
24            }
25        } else if (v != pre) low[u] = min(low[u], dis[v]);
26    }
27 }
28

```

```

29 int dep[N], par[N][20], cnt[N], id[N];
30 vector<int> bt[N];
31 void dfs1(int u, int pre = 0) {
32     dep[u] = dep[pre] + 1;
33     cnt[u] = cnt[pre] + art[u];
34     par[u][0] = pre;
35     for (int k = 1; k <= 18; k++) par[u][k] = par[par[u][k - 1]][k - 1];
36     for (auto v : bt[u]) if (v != pre) dfs1(v, u);
37 }
38
39 int lca(int u, int v) {
40     if (dep[u] < dep[v]) swap(u, v);
41     for (int k = 18; k >= 0; k--) if (dep[par[u][k]] >= dep[v])
42         u = par[u][k];
43     if (u == v) return u;
44     for (int k = 18; k >= 0; k--) if (par[u][k] != par[v][k]) u = par[u][k], v = par[v][k];
45     return par[u][0];
46 }
47
48 int dist(int u, int v) {
49     int lc = lca(u, v);
50     return cnt[u] + cnt[v] - 2 * cnt[lc] + art[lc];
51 }
52
53 int32_t main() {
54     ios_base::sync_with_stdio(0);
55     cin.tie(0);
56
57     int n, m;
58     cin >> n >> m;
59     while (m--) {
60         int u, v;
61         cin >> u >> v;
62         g[u].push_back(v);
63         g[v].push_back(u);
64     }
65     dfs(1);
66     for (int u = 1; u <= n; u++) {
67         if (bcc[u].size() > 1) { //AP
68             id[u] = ++sz;
69             art[id[u]] = 1; //if id in BCT is an AP on real graph
70             or not
71             for (auto v : bcc[u]) {
72                 bt[id[u]].push_back(v);
73                 bt[v].push_back(id[u]);
74             }
75         } else if (bcc[u].size() == 1) id[u] = bcc[u][0];
76     }
77     dfs1(1);
78     int q;
79     cin >> q;
80     while (q--) {
81         int u, v;
82         cin >> u >> v;
83         int ans;
84         if (u == v) ans = 0;
85         else ans = dist(id[u], id[v]) - art[id[u]] - art[id[v]];
86         cout << ans << '\n'; //number of articulation points in
87         //the path from u to v except u and v
88         //u and v are in the same bcc if ans == 0
89     }
90     return 0;

```

```

89 }
90 // https://codeforces.com/gym/102512/problem/A

```

## DSU ON Tree

```

1 // DSU on Tree (Sack)
2 // Computes for each node some info about its subtree
  efficiently
3 const int N = 200005;
4 vector<int> tree[N];
5 int col[N], sz[N];
6 bool big[N];
7 int cnt[N]; // frequency of colors
8 int cur_ans = 0; // example answer (can be modified)
9 int answer[N]; // answer for each node
10 // compute subtree sizes
11 void dfs_sz(int u, int p) {
12     sz[u] = 1;
13     for (int v : tree[u]) {
14         if (v == p) continue;
15         dfs_sz(v, u);
16         sz[u] += sz[v];
17     }
18 }
19 // add/remove contribution of subtree
20 void add(int u, int p, int val) {
21     cnt[col[u]] += val;
22     // example logic: maintain max frequency
23     cur_ans = max(cur_ans, cnt[col[u]]);
24     for (int v : tree[u]) {
25         if (v == p || big[v]) continue;
26         add(v, u, val);
27     }
28 }
29 // main DSU dfs
30 void dfs(int u, int p, bool keep) {
31     int mx = -1, heavy = -1;
32     for (int v : tree[u]) {
33         if (v != p && sz[v] > mx) {
34             mx = sz[v];
35             heavy = v;
36         }
37     }
38     // process light children
39     for (int v : tree[u]) {
40         if (v != p && v != heavy) dfs(v, u, false);
41     }
42     // process heavy child
43     if (heavy != -1) {
44         dfs(heavy, u, true); big[heavy] = true;
45     }
46     // add current node and light children
47     add(u, p, +1); answer[u] = cur_ans;
48     if (heavy != -1) big[heavy] = false;
49     // clear data if not keeping
50     if (!keep) {
51         add(u, p, -1); cur_ans = 0;
52     }
53 }

```

## DSU

```

1 void make_set(int v) {

```

```

2     parent[v] = v;
3 }
4 int find_set(int v) {
5     if (v == parent[v])
6         return v;
7     return parent[v] = find_set(parent[v]);
8 }
9 void union_sets(int a, int b) {
10     a = find_set(a);
11     b = find_set(b);
12     if (a != b)
13         parent[b] = a;
14 }

```

## LCA

```

1 void dfs(int u, int p) {
2     tree[u][0] = p;
3     dep[u] = dep[p] + 1;
4
5     for(int i = 1; i < B; i++) {
6         tree[u][i] = tree[tree[u][i - 1]][i - 1];
7     }
8     sz[u] = 1;
9     for(auto v : g[u]) {
10        if(v == p) continue;
11        dfs(v, u);
12        sz[u] += sz[v];
13    }
14 }
15
16 int lca(int u, int v) {
17     if(dep[u] > dep[v]) swap(u, v);
18     for(int i = B - 1; i >= 0; i--) {
19         if(dep[tree[v][i]] >= dep[u]) {
20             v = tree[v][i];
21         }
22     }
23
24     if(u == v) return u;
25     for(int i = B - 1; i >= 0; i--) {
26         if(tree[u][i] != tree[v][i]) {
27             u = tree[u][i];
28             v = tree[v][i];
29         }
30     }
31     return tree[u][0];
32 }

```

## euler\_path\_directed

```

1 vector<int> adj[200005];
2 int ptrs[200005];
3 int in_deg[200005], out_deg[200005];
4 vector<int> path; // node sequence
5 vector<int> edges; // edge sequence
6 int to[200005];
7 bool used[200005];
8
9 void dfs(int u) {
10     while(ptrs[u] < adj[u].size()) {
11         int e = adj[u][ptrs[u]++];
12         if(used[e]) continue;

```

```

13         used[e] = true;
14         dfs(to[e]);
15         edges.push_back(e);
16     }
17     path.push_back(u);
18 }
19
20
21
22     for(int i=0;i<m;i++){
23         int u,v;
24         cin >> u >> v;
25         adj[u].push_back(i);
26         to[i] = v;
27         out_deg[u]++;
28         in_deg[v]++;
29         if(i==0) s = u;
30     }
31
32     // degree check for Euler path
33     int start = -1, end = -1;
34     bool ok = true;
35
36     for(int i=0;i<n;i++){
37         int diff = out_deg[i] - in_deg[i];
38         if(diff == 1){
39             if(start != -1) ok = false;
40             start = i;
41         }
42         else if(diff == -1){
43             if(end != -1) ok = false;
44             end = i;
45         }
46         else if(diff != 0){
47             ok = false;
48         }
49     }
50
51     reverse(path.begin(), path.end());
52     reverse(edges.begin(), edges.end());

```

## kosaraju

```

1 typedef struct node {
2     int vis;
3     vector<int> adj;
4     vector<int> ulta;
5 } node;
6 node tree[300010];
7 vector<vector<int>> scc(300010);
8 stack<int> st;
9 int scount = 0;
10 void dfs(int u) {
11     tree[u].vis = 1;
12     f(i, tree[u].adj.size()) {
13         int next = tree[u].adj[i];
14         if(tree[next].vis==0) dfs(next);
15     }
16     st.push(u);
17 }
18 void dfs2(int u) {
19     // printf("%d %d\n", u, scount);
20     tree[u].vis = 1;
21     scc[scount-1].push_back(u);
22     f(i, tree[u].ulta.size()) {

```

```

23     int next = tree[u].ultra[i];
24     if(tree[next].vis==0) dfs2(next);
25 }
26 }

```

## mos on trees

```

1 struct Query { ll id, l, r, bno,k; }; //basic comparator
bool cmp(Query &a,Query &b) { if (a.bno != b.bno) {
return a.bno < b.bno; } else if (a.bno & 1) { return a
.r < b.r; } else { return a.r > b.r; } } ll n,q; ll bs
=700;// change is depending on problem Query qry[N];
ll a[N],b[N],st[N],en[N]; ll ans[N]; ll tt = -1; vll g
[N]; void gg(ll x, ll pr){ //trc(x); tt++; st[x] = tt;
a[tt] = x; //trc(tt,a[tt]); go(i,g[x]){ if(i == pr)
cu; gg(i,x); } tt++; en[x] = tt; a[tt] = x; } const ll
LG = 30; ll tim=0; ll parent[LG][N]; ll tin[N], tout[
N], level[N]; void dfs(ll k, ll par, ll lvl) { tin[k
]=++tim; parent[0][k]=par; level[k]=lvl; for(auto it:g
[k]) { if(it==par) continue; dfs(it, k, lvl+1); } tout
[k]=tim; } ll walk(ll u, ll h) { for(ll i=LG-1;i>=0;i
--){ if((h>>i) & 1) u = parent[i][u]; } return u; }
void precompute() { for(ll i=1;i<LG;i++) for(ll j=1;j
<=n;j++) if(parent[i-1][j]) parent[i][j]=parent[i-1][
parent[i-1][j]]; } ll lca(ll u, ll v) { if(level[u]<
level[v]) swap(u,v); ll diff=level[u]-level[v]; for(ll
i=LG-1;i>=0;i--) { if((1<<i) & diff) { u=parent[i][u
]; } } if(u==v) return u; for(ll i=LG-1;i>=0;i--) { if
(parent[i][u] && parent[i][u]!=parent[i][v]) { u=
parent[i][u]; v=parent[i][v]; } } return parent[0][u];
} ll dist(ll u, ll v) { return level[u] + level[v] -
2 * level[lca(u, v)]; } ll lp,rp,cnt; ll freq
[1000001]; ll freq2[1000005]; //unordered_map<ll,ll>
freq,freq2; void add(ll val) { freq[val]++; freq2[b[
val]]++; if(freq[val] == 2){ freq2[b[val]]-=2; // if(
freq2[b[val]] == 0){ // cnt--; // } } else{ // if(
freq2[b[val]] == 1) cnt++; } } void remove(ll val) {
freq[val]--; freq2[b[val]]--; if(freq[val] == 1){
freq2[b[val]]+=2; // if(freq2[b[val]] == 1){ // cnt++;
// } } else{ // if(freq2[b[val]] == 0) cnt--; } } ll
get_answer() { return cnt; } void solve() { cin >> n>>
q; //trc(n,q); mll m; ll dd = 0; for (ll i = 1; i <= n
; i++) { cin >> b[i]; if(!m.count(b[i])){ m[b[i]] = dd
++; } b[i] = m[b[i]]; } f0(n-1){ ipp(x,y); //trc(x,y);
g[x].pb(y); g[y].pb(x); } gg(1,0); dfs(1,0,0);
precompute(); for (ll i = 0; i < q; i++) { ll x, y;
cin >> x >> y; if(level[x]>level[y]){ swap(x,y); } ll
p = lca(x,y); // trc(x,y,p); ll l,r; if(p == x){ l =
st[x]; r = st[y]; qry[i].k = -1; } else{ if(st[x]>st[y]
) swap(x,y); l = en[x]; r = st[y]; qry[i].k = st[p];
// trc(l,r,p,st[p]); // f(j,l,r+1){ // trc(j,a[j]); //
} } qry[i].id = i; qry[i].l = l; qry[i].r = r; qry[i
].bno = l / bs; } // sort the queries sort(qry, qry +
q, cmp); // initialise the data-structure lp = 0; rp =
-1; cnt = 0; // add(a[0]); // answer the queries for
(ll i = 0; i < q; i++) { ll l = qry[i].l; ll r = qry[i
].r; while (rp < r) { add(a[+rp]); } while (lp > l) {
add(a[--lp]); } while (rp > r) { remove(a[rp--]); }
while (lp < l) { remove(a[lp++]); } ans[qry[i].id] =
get_answer(); if(qry[i].k != -1){ // if(freq2[b[a[qry[
i].k]]] == 0){ // ans[qry[i].id]++; // } } } for (ll i
= 0; i < q; i++) { cout << ans[i] << endl; } } //
this require the element to be added on both side l,r
and removed l,r // but we can also use mo's when data
structure allow the removal from the side where we

```

```

added like rollback dsu void fun() { solve(); }
int32_t main() { // tasklist // taskkill /f /im a.exe
depressed(); ll tc1 = 1; //cin>>tc1; while(tc1--){
fun(); } //cer; return 0; }

```

## roll\_dsu

```

1 struct DSU { int connected; int sz[N]; pair<int, int> par[N]
}; //stores <parent, parity of distance from parent>;
vector<int> changes; void init(int n) { for(int i=1;i
<=n;i++) { par[i]={i, 0}; sz[i]=1; } connected=n; }
pair<int, int> getPar(int x) { if(par[x].first==x)
return par[x]; pair<int, int> p = getPar(par[x].first)
; p.second ^= par[x].second; return p; } bool unite(
int u, int v) //Returns 1 if the graph is no longer
bipartite { pair<int, int> x=getPar(u), y=getPar(v);
if(x.first==y.first) { if(x.second==y.second) return
1; changes.push_back(-1); return 0; } connected--; if(
sz[x.first] < sz[y.first]) swap(x, y); par[y.first]={x
.first, 1^x.second^y.second}; sz[x.first]+=sz[y.first
]; changes.push_back(y.first); return 0; } void undo()
//Reversed the previous performed operation { if(!
changes.size()) return; if(changes.back()!==-1) { int x
=changes.back(); sz[par[x].first]-=sz[x]; par[x]={x,
0}; } changes.pop_back(); } };

```

## Number Theory Math

### CRT

```

1 __int128 ExU(__int128 a, __int128 b, __int128 &x, __int128
&y) {
2     if(b == 0) {
3         x = 1;
4         y = 0;
5         return a;
6     }
7     __int128 x1, y1;
8     __int128 d = ExU(b, a%b, x1, y1);
9     x = y1;
10    y = x1 - (a/b)*y1;
11    return d;
12 }
13 void solve() {
14     cout << "Case " << ii++ << ": ";
15     int n; cin >> n;
16     vector<int> a(n), m(n);
17     for (int i = 0; i < n; i++) cin >> m[i] >> a[i];
18     int a1 = a[0], m1 = m[0];
19     for (int i = 1; i < n; i++) {
20         __int128 a2 = a[i], m2 = m[i];
21         __int128 x, y;
22         __int128 g = ExU(m1, m2, x, y);
23         if((a1-a2) % g != 0) {
24             cout << "Impossible" << endl;
25             return;
26         }
27         __int128 k = (a1-a2) / g, lcm = (m1/g) * m2;
28         __int128 a = a1 - m1*(k * x);
29         a1 = (a % lcm);
30         m1 = lcm;
31     }

```

```

32 }
33 cout << a1 << endl;
34 }

```

## Diophantile

```

1 int gcd(int a, int b, int& x, int& y) {
2     if (b == 0) { x = 1; y = 0; return a;
3     }
4     int x1, y1;
5     int d = gcd(b, a % b, x1, y1);
6     x = y1; y = x1 - y1 * (a / b);
7     return d;
8 }
9
10 bool find_any_solution(int a, int b, int c, int &x0, int &
y0, int &g) {
11     g = gcd(abs(a), abs(b), x0, y0);
12     if (c % g) {
13         return false;
14     }
15     x0 *= c / g; y0 *= c / g;
16     if (a < 0) x0 = -x0;
17     if (b < 0) y0 = -y0;
18     return true;
19 }
20 void shift_solution(int &x, int &y, int a, int b, int cnt
) {
21     x += cnt * b; y -= cnt * a;
22 }
23 int find_all_solutions(int a, int b, int c, int minx, int
maxx, int miny, int maxy) {
24     int x, y, g;
25     if (!find_any_solution(a, b, c, x, y, g)) return 0;
26     a /= g; b /= g;
27     int sign_a = a > 0 ? +1 : -1;
28     int sign_b = b > 0 ? +1 : -1;
29     shift_solution(x, y, a, b, (minx - x) / b);
30     if (x < minx) shift_solution(x, y, a, b, sign_b);
31     if (x > maxx) return 0;
32     int lx1 = x;
33     shift_solution(x, y, a, b, (maxx - x) / b);
34     if (x > maxx) shift_solution(x, y, a, b, -sign_b);
35     int rx1 = x; shift_solution(x, y, a, b, -(miny - y) / a
);
36     if (y < miny) shift_solution(x, y, a, b, -sign_a);
37     if (y > maxy) return 0;
38     int lx2 = x;
39     shift_solution(x, y, a, b, -(maxy - y) / a);
40     if (y > maxy) shift_solution(x, y, a, b, sign_a);
41     int rx2 = x;
42     if (lx2 > rx2) swap(lx2, rx2);
43     int lx = max(lx1, lx2);
44     int rx = min(rx1, rx2);
45     if (lx > rx) return 0;
46     return (rx - lx) / abs(b) + 1;
47 }

```

## FFT

```

1 const int N = 3e5 + 9;
2 const double PI = acos(-1);
3 struct base {

```

```

4 double a, b;
5 base(double a = 0, double b = 0) : a(a), b(b) {}
6 const base operator + (const base &c) const
7 { return base(a + c.a, b + c.b); }
8 const base operator - (const base &c) const
9 { return base(a - c.a, b - c.b); }
10 const base operator * (const base &c) const
11 { return base(a * c.a - b * c.b, a * c.b + b * c.a); }
12 };
13 void fft(vector<base> &p, bool inv = 0) {
14     int n = p.size(), i = 0;
15     for(int j = 1; j < n - 1; ++j) {
16         for(int k = n >> 1; k > (i ^= k); k >>= 1);
17         if(j < i) swap(p[i], p[j]);
18     }
19     for(int l = 1, m; (m = l << 1) <= n; l <== 1) {
20         double ang = 2 * PI / m;
21         base wn = base(cos(ang), (inv ? 1. : -1.) * sin(ang)), w =
22         ;
23         for(int i = 0, j, k; i < n; i += m) {
24             for(w = base(1, 0), j = i, k = i + 1; j < k; ++j, w =
25             w * wn) {
26                 base t = w * p[j + 1];
27                 p[j + 1] = p[j] - t;
28                 p[j] = p[j] + t;
29             }
30         }
31     }
32     if(inv) for(int i = 0; i < n; ++i) p[i].a /= n, p[i].b /=
33     n;
34     vector<long long> multiply(vector<int> &a, vector<int> &b)
35     {
36         int n = a.size(), m = b.size(), t = n + m - 1, sz = 1;
37         while(sz < t) sz <== 1;
38         vector<base> x(sz), y(sz), z(sz);
39         for(int i = 0; i < sz; ++i) {
40             x[i] = i < (int)a.size() ? base(a[i], 0) : base(0, 0);
41             y[i] = i < (int)b.size() ? base(b[i], 0) : base(0, 0);
42         }
43         fft(x), fft(y);
44         for(int i = 0; i < sz; ++i) z[i] = x[i] * y[i];
45         fft(z, 1);
46         vector<long long> ret(sz);
47         for(int i = 0; i < sz; ++i) ret[i] = (long long) round(z[
48         i].a);
49         while((int)ret.size() > 1 && ret.back() == 0) ret.
50         pop_back();
51         return ret;
52     }
53 }

```

## LSieveBigModTernaryS

```

1 int a[], n elements;
2 int mx = 0; int ans = 0;
3 for (int i = 0; i < n; i++) {
4     mx = max(mx+a[i], (int)0);
5     ans = max(ans, mx);
6 }
7 #define sz 1000
8 vector<int> prime;
9 bitset<sz> bs;
10 void sieve() {
11     bs.set(); // all bit = 1..
12     bs[0] = bs[1] = 0;

```

```

13 prime.push_back(2);
14 for (int i = 3; i <= sz; i += 2) {
15     if(bs[i]) {
16         for (int j = i*i; j <= sz; j += i)
17             bs[j] = 0;
18         prime.push_back(i);
19     }
20 }
21 }
22 void solve()
23 {
24     int n = 12;
25     int a[] = {60, 70, 80, 90, 100, 200, 400, 500, 600, 700,
26     900, 1000};
27     int r = n-1; int l = 0;
28     // cout << (r-l) / 2;
29     while(l < r) {
30         int c = (r-l) / 3;
31         int m1 = l + c;
32         int m2 = r - c;
33         // cout << a[m1] << ' ' << a[m2] << endl;
34         if(a[m1] < a[m2]) {
35             r = m2 - 1;
36         } else if(a[m1] > a[m2]) {
37             l = m1 + 1;
38         } else {
39             l = m1 + 1; r = m2 - 1;
40         }
41     } cout << a[l] << endl;
42 }
43 int mod = 1e9 + 7;
44 // linear sieve.
45 vector<int> prime;
46 bool is_composite[MAXN];
47 void sieve (int n) {
48     std::fill (is_composite, is_composite + n, false);
49     for (int i = 2; i < n; ++i) {
50         if (!is_composite[i]) prime.push_back (i);
51         for (int j = 0; j < prime.size () && i * prime[j] < n;
52             ++j) {
53             is_composite[i * prime[j]] = true;
54             if (i % prime[j] == 0) break;
55         }
56     }
57 }

```

## Miller Rabin

```

1 using u64 = uint64_t;
2 using u128 = __uint128_t;
3 u64 binpower(u64 base, u64 e, u64 mod) {
4     u64 result = 1;
5     base %= mod;
6     while (e) {
7         if (e & 1)
8             result = (u128)result * base % mod;
9         base = (u128)base * base % mod;
10        e >>= 1;
11    }
12    return result;
13 }
14 bool check_composite(u64 n, u64 a, u64 d, int s) {
15     u64 x = binpower(a, d, n);
16     if (x == 1 || x == n - 1)

```

```

17     return false;
18     for (int r = 1; r < s; r++) {
19         x = (u128)x * x % n;
20         if (x == n - 1)
21             return false;
22     }
23     return true;
24 };
25 bool MillerRabin(u64 n, int iter=5) { // returns true if n
26     // is probably prime, else returns false.
27     if (n < 4)
28         return n == 2 || n == 3;
29     int s = 0;
30     u64 d = n - 1;
31     while ((d & 1) == 0) {
32         d >>= 1;
33         s++;
34     }
35     for (int i = 0; i < iter; i++) {
36         int a = 2 + rand() % (n - 3);
37         if (check_composite(n, a, d, s))
38             return false;
39     }
40     return true;
41 }
42 bool MillerRabin(u64 n) { // returns true if n is prime,
43     // else returns false.
44     if (n < 2)
45         return false;
46     int r = 0;
47     u64 d = n - 1;
48     while ((d & 1) == 0) {
49         d >>= 1;
50         r++;
51     }
52     for (int a : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31,
53     37}) {
54         if (n == a)
55             return true;
56         if (check_composite(n, a, d, r))
57             return false;
58     }
59     return true;
60 }

```

## NTT

```

1 // Common prime for NTT: 998244353 = 119 * 2^23 + 1
2 // Primitive root: 3
3 const int MOD = 998244353;
4 const int G = 3;
5 // Number Theoretic Transform
6 // on: true for Inverse NTT, false for Forward NTT
7 void ntt(vector<long long> &a, bool invert) {
8     int n = a.size();
9
10    // Bit-reversal permutation
11    for (int i = 1, j = 0; i < n; i++) {
12        int bit = n >> 1;
13        for (; j & bit; bit >>= 1)
14            j ^= bit;
15        j ^= bit;
16        if (i < j) swap(a[i], a[j]);
17    }
18    // Butterfly operations

```

```

19 for (int len = 2; len <= n; len <= 1) {
20     long long wlen = power(G, (MOD - 1) / len);
21     if (invert) wlen = modInverse(wlen);
22
23     for (int i = 0; i < n; i += len) {
24         long long w = 1;
25         for (int j = 0; j < len / 2; j++) {
26             long long u = a[i + j], v = (a[i + j + len / 2]
27                 * w) % MOD;
28             a[i + j] = (u + v) < MOD ? (u + v) : (u + v -
29                 MOD);
30             a[i + j + len / 2] = (u - v >= 0) ? (u - v) : (
31                 u - v + MOD);
32             w = (w * wlen) % MOD;
33         }
34     }
35     // Scale results for Inverse NTT
36     if (invert) {
37         long long n_inv = modInverse(n);
38         for (long long x : a)
39             x = (x * n_inv) % MOD;
40     }
41 }
42 // Polynomial Multiplication
43 vector<long long> multiply(vector<long long> const& a,
44     vector<long long> const& b) {
45     vector<long long> fa(a.begin(), a.end()), fb(b.begin(),
46         b.end());
47     int n = 1;
48     while (n < a.size() + b.size())
49         n <<= 1;
50     fa.resize(n);
51     fb.resize(n);
52     ntt(fa, false);
53     ntt(fb, false);
54     for (int i = 0; i < n; i++)
55         fa[i] = (fa[i] * fb[i]) % MOD;
56     ntt(fa, true);
57     // Optional: Resize to actual degree (removes trailing
58         zeros)
59     // while (fa.size() > 1 && fa.back() == 0) fa.pop_back()
60     ;
61     return fa;
62 }

```

## ballot

```

1 // Returns the number of paths from (0,0) to (a,b) staying
2   at or below y = x + k
3 long long ballot_with_offset(int a, int b, int k) {
4     if (b > a + k) return 0; // Already invalid
5     long long total = nCr(a + b, a);
6     long long invalid = nCr(a + b, a + k + 1);
7     return (total - invalid + mod) % mod;
8 }
9 // Non-strict Ballot Number: A >= B at all prefixes
10 long long ballot_non_strict(int a, int b) {
11     return ballot_with_offset(a, b, 0);
12 }
13
14 // Strict Ballot Number: A > B at all non-empty prefixes
15 long long ballot_strict(int a, int b) {
16     if (a <= b) return 0;

```

```

17     return ballot_non_strict(a - 1, b);
18 }

```

## matexpo

```

1 vector<vector<ll>> matmultiply(vector<vector<ll>>& a,
2     vector<vector<ll>>& b) {
3     if (a[0].size() != b.size()) return vector<vector<ll>>(0)
4     ;
5     else {
6         vector<vector<ll>> ans(a.size(), vector<ll>(b[0].size()
7             (0,0));
8         f(i,a.size()) {
9             f(j,b[0].size()) {
10                 f(k,b.size()) ans[i][j] = (ans[i][j] + ((a[i][k]*
11                     b[k][j])%mod))%mod;
12             }
13         }
14         return ans;
15     }
16 }
17 vector<vector<ll>> matexpo(vector<vector<ll>>& a, ll exp) {
18     vector<vector<ll>> res(a[0].size(), vector<ll>(a[0].size()
19         (0,0));
20     f(i,a[0].size()) res[i][i] = 1;
21     while (exp > 0) {
22         if (exp & 1) res = matmultiply(res, a);
23         a = matmultiply(a, a);
24         exp /= 2;
25     }
26     return res;
27 }
28 int main() {
29     ll n;
30     scanf("%lld", &n);
31     if (n == 0) printf("0\n");
32     else {
33         vector<vector<ll>> arr(2, vector<ll>(2, 1));
34         arr[1][1] = 0;
35         arr = matexpo(arr, n);
36         printf("%lld\n", arr[0][1]);
37     }
38     return 0;
39 }

```

## mobius

```

1 int mob[N];
2 void mobius() {
3     mob[1] = 1;
4     for (int i = 2; i < N; i++) {
5         mob[i]--;
6         for (int j = i + i; j < N; j += i) {
7             mob[j] -= mob[i];
8         }
9     }
10 }
11 bool vis[N];
12 vector<int> d[N];
13 int mul[N];
14 void add(int x, int k) {
15     for (auto y: d[x]) {

```

```

16         mul[y] += k;
17     }
18 }
19 int query(int x) {
20     int ans = 0;
21     for (auto y: d[x]) {
22         ans += mul[y] * mob[y];
23     }
24     return ans;
25 }

```

## ncr

```

1 // nCr: (n,0)=1, (n,1)=n, (n,r)=(n-1,r)+(n-1,r-1)
2 // Circular permutation: nPr/r
3 ll fact[MAX], invFact[MAX];
4 // nCr_mod_p_by_lucas_for_small_p._O(LogN)
5 ll Lucas(ll n, ll m) {
6     if (n == 0 && m == 0) return 1ll;
7     ll ni = n % MOD;
8     ll mi = m % MOD;
9     if (mi > ni) return 0;
10    return (Lucas(n / MOD, m / MOD) * nCr(ni, mi)) % MOD;
11 }
12 ll nCr_by_lucas(ll n, ll r) { return Lucas(n, r); }
13 // nCr_mod_M, M_is_not_prime
14 const int N = 142858; // need primes <=M (nCr%M)
15 int spf[N];
16 vector<int> primes;
17 void sieve() {
18     for (int i = 2; i < N; i++) {
19         if (spf[i] == 0) spf[i] = i, primes.push_back(i);
20         int sz = primes.size();
21         for (int j = 0; j < sz && i * primes[j] < N && primes[j]
22             <= spf[i]; j++) spf[i * primes[j]] = primes[j];
23     }
24 }
25 // returns n! % mod, mod = multiple of p
26 // O(mod * log(n))
27 int factmod(ll n, int p, const int mod) {
28     vector<int> f(mod + 1);
29     f[0] = 1 % mod;
30     for (int i = 1; i <= mod; i++) {
31         if (i % p) f[i] = 1LL * f[i - 1] * i % mod;
32         else f[i] = f[i - 1];
33     }
34     int ans = 1 % mod;
35     while (n > 1) {
36         ans = 1LL * ans * f[n % mod] % mod;
37         ans = 1LL * ans * BigMod(f[mod], n / mod, mod) % mod;
38         n /= p;
39     }
40     return ans;
41 }
42 ll multiplicity(ll n, int p) {
43     ll ans = 0;
44     while (n) {
45         n /= p;
46         ans += n;
47     }
48     return ans;
49 }
50 // C(n, r) modulo p^k
51 // O(p^k log n)

```

```

51 int ncr(ll n, ll r, int p, int k){
52     if (n < r || r < 0) return 0;
53     int mod = 1;
54     for (int i = 0; i < k; i++) { mod *= p; }
55     ll t = multiplicity(n, p) - multiplicity(r, p) -
        multiplicity(n - r, p);
56     if (t >= k) return 0;
57     int ans = 1LL * factmod(n, p, mod) * inverse(factmod(r,
        p, mod), mod) % mod * inverse(factmod(n - r, p, mod
        ), mod) % mod;
58     ans = 1LL * ans * BigMod(p, t, mod) % mod;
59     return ans;
60 }
61 pair<ll, ll> CRT(ll a1, ll m1, ll a2, ll m2){
62     ll p, q; ll g = e_gcd(m1, m2, p, q);
63     if (a1 % g != a2 % g) return make_pair(0, -1);
64     ll m = m1 / g * m2;
65     p = (p % m + m) % m;
66     q = (q % m + m) % m;
67     return make_pair((p * a2 % m * (m1 / g) % m + q * a1 % m
        * (m2 / g) % m) % m, m);
68 }
69 // O(m log(n) log(m))
70 int ncr(ll n, ll r, int m){
71     if (n < r || r < 0) return 0;
72     pair<ll, ll> ans({0, 1});
73     while (m > 1){
74         int p = spf[m], k = 0, cur = 1;
75         while (m % p == 0){
76             m /= p; cur *= p; ++k;
77         }
78         ans = CRT(ans.first, ans.second, ncr(n, r, p, k), cur
            );
79     }
80     return ans.first;
81 }

```

## stirling\_number

```

1 // S(n,k) = (1/k!) * sum_{i=0..k} (-1)^i * C(k,i) * (k-i)^n
2 int main() {
3     int n, k; cin >> n >> k;
4     vector<ll> fact(k+1, 1), invfact(k+1, 1);
5     ll ans = 0;
6     for (int i = 0; i <= k; i++) {
7         ll term = fact[k] * invfact[i] % MOD * invfact[k-i] %
            MOD;
8         term = term * power(k - i, n) % MOD;
9         if (i & 1) ans = (ans - term + MOD) % MOD;
10        else ans = (ans + term) % MOD;
11    }
12    ans = ans * modinv(fact[k]) % MOD;
13    cout << ans << "\n";
14 }

```

## totient

```

1 int phi(int n) {
2     int result = n;
3     for (int i = 2; i * i <= n; i++) {
4         if (n % i == 0) {
5             while (n % i == 0) n /= i;
6             result -= result / i;

```

```

7         }
8     }
9     if (n > 1) result -= result / n;
10    return result;
11 }
12 void phi_1_to_n(int n) {
13     vector<int> phi(n + 1);
14     for (int i = 0; i <= n; i++) phi[i] = i;
15     for (int i = 2; i <= n; i++) {
16         if (phi[i] == i) {
17             for (int j = i; j <= n; j += i) phi[j] -= phi[j] /
                i;
18         }
19     }
20 }
21 void phi_1_to_n(int n) {
22     vector<int> phi(n + 1);
23     phi[0] = 0; phi[1] = 1;
24     for (int i = 2; i <= n; i++) phi[i] = i - 1;
25
26     for (int i = 2; i <= n; i++)
27         for (int j = 2 * i; j <= n; j += i) phi[j] -= phi[i];
28 }

```

## String

### AhoCorasick

```

1 typedef struct AC {
2     static const int K = 26;
3     struct node {
4         int nxt[K], link, leaf, par;
5         vector<int> ind;
6         char pch;
7         node(int par = -1, char pch = '$') : par(par), pch(
            pch) {
8             fill(begin(nxt), end(nxt), -1);
9             leaf = false;
10            link = -1;
11        }
12    };
13    vector<node> t;
14    vector<vector<int>> ad;
15    vector<int> st, en, mp;
16    vector<int> cnt;
17    int dt = 0;
18    AC() {
19        mp = vector<int>(1<<8, 0);
20        for (char ch = 'a'; ch <= 'z'; ++ch) mp[ch] = ch - 'a
            ';
21        t.resize(1);
22        cnt.resize(1);
23    }
24    int add(string str, int i) {
25        int ptr = 0;
26        for (auto ch : str) {
27            if (t[ptr].nxt[mp[ch]] < 0) {
28                t[ptr].nxt[mp[ch]] = t.size();
29                t.push_back(node(ptr, ch));
30                cnt.push_back(0);
31            }
32            ptr = t[ptr].nxt[mp[ch]];
33        }
34        t[ptr].leaf = true;

```

```

35        t[ptr].ind.push_back(i);
36        return ptr;
37    }
38    int get_link(int v) {
39        if (t[v].link == -1) {
40            t[v].link = (!v || !t[v].par) ? 0 : go(get_link(t[
                v].par), t[v].pch);
41        }
42        return t[v].link;
43    }
44    int go(int v, char ch) {
45        if (t[v].nxt[mp[ch]] < 0) {
46            t[v].nxt[mp[ch]] = !v ? 0 : go(get_link(v), ch);
47        }
48        return t[v].nxt[mp[ch]];
49    }
50    void dfs(int v) {
51        st[v] = ++dt;
52        for (int u : ad[v]) {
53            dfs(u);
54            cnt[v] += cnt[u];
55        }
56        en[v] = dt;
57    }
58    void calc() {
59        int k = t.size();
60        st.resize(k);
61        en.resize(k);
62        ad.resize(k);
63        for (int i = 1; i < k; i++)
64            ad[get_link(i)].push_back(i);
65        dfs(0);
66    }
67 } cc;
68 int cs = 1;
69 void solve() {
70     cc.bc;
71     int n; cin >> n;
72     vector<int> res(n + 1, 0);
73     string text;
74     cin >> text;
75     string p;
76     for (int i = 1; i <= n; i++) {
77         cin >> p;
78         bc.add(p, i);
79     }
80     int vertex = 0;
81     for (int i = 0; i < text.length(); i++) {
82         vertex = bc.go(vertex, text[i]);
83         bc.cnt[vertex]++;
84     }
85     bc.calc();
86     for (int i = 1; i < bc.t.size(); i++) {
87         for (int &x : bc.t[i].ind) res[x] += bc.cnt[i];
88     }
89     cout << "Case " << cs++ << ":\n";
90     for (int i = 1; i <= n; i++) cout << res[i] << "\n";
91 }

```

## Manacher

```

1 struct Manacher {
2     vector<int> p[2];
3     // p[1][i] = (max odd length palindrome centered at i) /
        2 [floor division]

```

```

4 // p[0][i] = same for even, it considers the right center
5 // e.g. for s = "abbabba", p[1][3] = 3, p[0][2] = 2
6 Manacher(string s) {
7     int n = s.size();
8     p[0].resize(n + 1);
9     p[1].resize(n);
10    for (int z = 0; z < 2; z++) {
11        for (int i = 0, l = 0, r = 0; i < n; i++) {
12            int t = r - i + !z;
13            if (i < r) p[z][i] = min(t, p[z][l + t]);
14            int L = i - p[z][i], R = i + p[z][i] - !z;
15            while (L >= 1 && R + 1 < n && s[L - 1] == s[R + 1])
16                p[z][i]++, L--, R++;
17            if (R > r) l = L, r = R;
18        }
19    }
20 }
21 bool is_palindrome(int l, int r) {
22     int mid = (l + r + 1) / 2, len = r - l + 1;
23     return 2 * p[len % 2][mid] + len % 2 >= len;
24 }
25 };
26 int32_t main() {
27     string s; cin >> s;
28     Manacher M(s);
29     int n = s.size();
30     for (int i = 0; i < n; i++) {
31         cout << 2 * M.p[1][i] + 1 << ' ';
32         if (i + 1 < n) cout << 2 * M.p[0][i + 1] << ' ';
33     }
34     cout << '\n';
35     return 0;
36 }

```

## Suffix Automata

```

1 // Suffix Automaton (SAM)
2 // Handles all substrings of a string in linear time
3 struct State {
4     int next[26];
5     int link, len;
6     State() {
7         memset(next, -1, sizeof next);
8         link = -1;
9         len = 0;
10    }
11 };
12 vector<State> st;
13 int last;
14 // initialize
15 void sa_init() {
16     st.clear(); st.push_back(State()); last = 0;
17 }
18 // extend by character c
19 void sa_extend(char ch) {
20     int c = ch - 'a';
21     int cur = st.size();
22     st.push_back(State());
23     st[cur].len = st[last].len + 1;
24     int p = last;
25     while (p != -1 && st[p].next[c] == -1) {
26         st[p].next[c] = cur;
27         p = st[p].link;
28     }
29     if (p == -1) {

```

```

30         st[cur].link = 0;
31     } else {
32         int q = st[p].next[c];
33         if (st[p].len + 1 == st[q].len) {
34             st[cur].link = q;
35         } else {
36             int clone = st.size();
37             st.push_back(st[q]);
38             st[clone].len = st[p].len + 1;
39             while (p != -1 && st[p].next[c] == q) {
40                 st[p].next[c] = clone; p = st[p].link;
41             }
42             st[q].link = st[cur].link = clone;
43         }
44     }
45     last = cur;
46 }
47 // count distinct substrings
48 long long count_distinct() {
49     long long ans = 0;
50     for (auto &s : st)
51         ans += s.len - (s.link == -1 ? 0 : st[s.link].len);
52     return ans;
53 }

```

## palindromitree

```

1 const int MAXN = 105000;
2 struct node {
3     int next[26];
4     int len;
5     int sufflink;
6     int num;
7 };
8 int len;
9 char s[MAXN];
10 node tree[MAXN];
11 int num; // node 1 - root with len -1, node 2 - root with len 0
12 int suff; // max suffix palindrome
13 long long ans;
14 bool addLetter(int pos) {
15     int cur = suff, curlen = 0;
16     int let = s[pos] - 'a';
17     while (true) {
18         curlen = tree[cur].len;
19         if (pos - 1 - curlen >= 0 && s[pos - 1 - curlen] == s[pos])
20             break;
21         cur = tree[cur].sufflink;
22     }
23     if (tree[cur].next[let]) {
24         suff = tree[cur].next[let];
25         return false;
26     }
27     num++;
28     suff = num;
29     tree[num].len = tree[cur].len + 2;
30     tree[cur].next[let] = num;
31     if (tree[num].len == 1) {
32         tree[num].sufflink = 2;
33         tree[num].num = 1;
34         return true;
35     }
36     while (true) {

```

```

37         cur = tree[cur].sufflink;
38         curlen = tree[cur].len;
39         if (pos - 1 - curlen >= 0 && s[pos - 1 - curlen] == s[pos]) {
40             tree[num].sufflink = tree[cur].next[let];
41             break;
42         }
43     }
44     tree[num].num = 1 + tree[tree[num].sufflink].num;
45     return true;
46 }
47 void initTree() {
48     num = 2; suff = 2;
49     tree[1].len = -1; tree[1].sufflink = 1;
50     tree[2].len = 0; tree[2].sufflink = 1;
51 }
52 int main() {
53     //assert(freopen("input.txt", "r", stdin));
54     //assert(freopen("output.txt", "w", stdout));
55     gets(s); len = strlen(s);
56     initTree();
57     for (int i = 0; i < len; i++) {
58         addLetter(i);
59         ans += tree[suff].num;
60     }
61 }

```

## suffixArray

```

1 struct SuffixArray {
2     int n; string s;
3     vector<int> sa, rankv, tmp, lcp;
4     SuffixArray(const string &str) {
5         s = str; n = s.size(); sa.resize(n); rankv.resize(n);
6         tmp.resize(n); build(); build_lcp();
7     }
8     void counting_sort(int k) {
9         int maxi = max(256ll, (long long)n) + 5;
10        vector<int> cnt(maxi, 0), sa2(n);
11        for (int i = 0; i < n; i++) {
12            int key = (i + k < n) ? rankv[i + k] + 1 : 0;
13            cnt[key]++;
14        }
15        for (int i = 1; i < maxi; i++) cnt[i] += cnt[i - 1];
16        for (int i = n - 1; i >= 0; i--) {
17            int key = (sa[i] + k < n) ? rankv[sa[i] + k] + 1 : 0;
18            sa2[--cnt[key]] = sa[i];
19        }
20        sa.swap(sa2);
21    }
22    void build() {
23        for (int i = 0; i < n; i++) {
24            sa[i] = i; rankv[i] = (unsigned char)s[i];
25        }
26        for (int k = 1; k < n; k <= 1) {
27            counting_sort(k);
28            counting_sort(0);
29            tmp[sa[0]] = 0;
30            int r = 0;
31            for (int i = 1; i < n; i++) {
32                int curl = rankv[sa[i]], cur2 = (sa[i] + k < n ? rankv[sa[i] + k] : -1);
33                int prev1 = rankv[sa[i - 1]], prev2 = (sa[i - 1] + k < n ? rankv[sa[i - 1] + k] : -1);

```

```

33         if (cur1 != prev1 || cur2 != prev2) r++;
34         tmp[sa[i]] = r;
35     }
36     rankv.swap(tmp);
37     if (r == n - 1) break;
38 }
39 }
40 void build_lcp() {
41     lcp.assign(max(0, n - 1), 0);
42     vector<int> inv(n);
43     for (int i = 0; i < n; ++i) inv[sa[i]] = i;
44     int k = 0;
45     for (int i = 0; i < n; ++i) {
46         if (inv[i] == n - 1) { k = 0; continue; }
47         int j = sa[inv[i] + 1];
48         while (i + k < n && j + k < n && s[i + k] == s[j + k]) ++k;
49         lcp[inv[i]] = k;
50         if (k) --k;
51     }
52 }
53 };
54 void solve(const string &s, const string &t) {
55     // build local combined string so we don't mutate inputs
56     int n = s.size();
57     string combined = s + '$' + t;
58     int N = combined.size();
59     SuffixArray sa(combined);
60 }

```

## z kmp

```

1 vector<int> prefix_function(string s) {
2     int n = (int)s.length();
3     vector<int> pi(n);
4     for (int i = 1; i < n; i++) {
5         int j = pi[i-1];
6         while (j > 0 && s[i] != s[j])
7             j = pi[j-1];
8         if (s[i] == s[j])
9             j++;
10        pi[i] = j;
11    }
12    return pi;
13 }
14 vector<int> z_function(string pattern, string text) {
15     string s = pattern + "$" + text;
16     int n = size(s);
17     vector<int> z(n);
18     for (int i = 1, l = 0, r = 0; i < n; ++i) {
19         if (i <= r)
20             z[i] = min (r - i + 1, z[i - l]);
21         while (i + z[i] < n && s[z[i]] == s[i + z[i]])
22             ++z[i];
23         if (i + z[i] - 1 > r)
24             l = i, r = i + z[i] - 1;
25     }
26     return z;
27 }

```